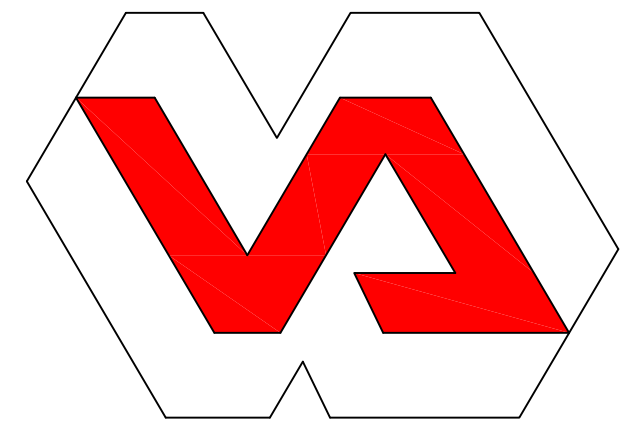
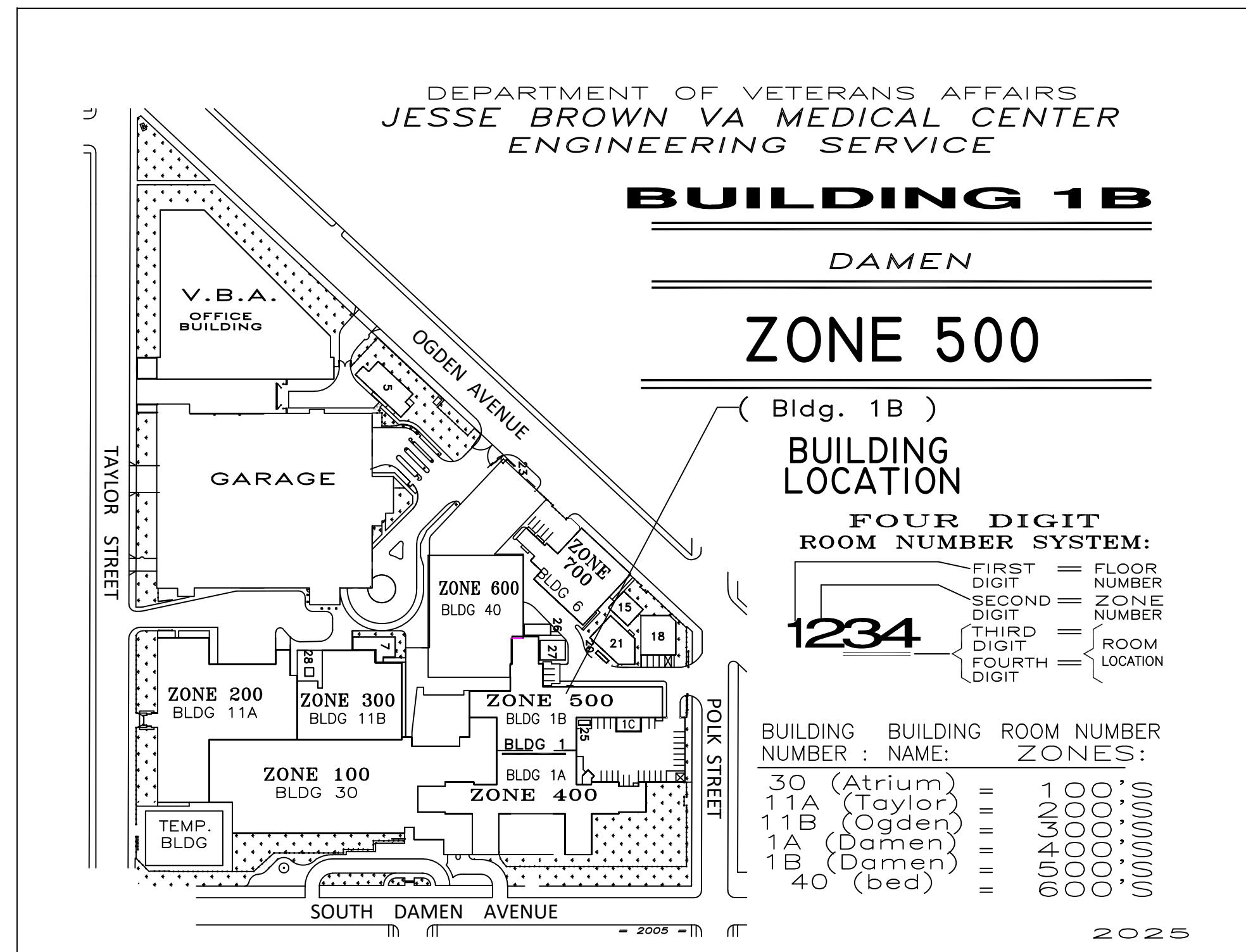


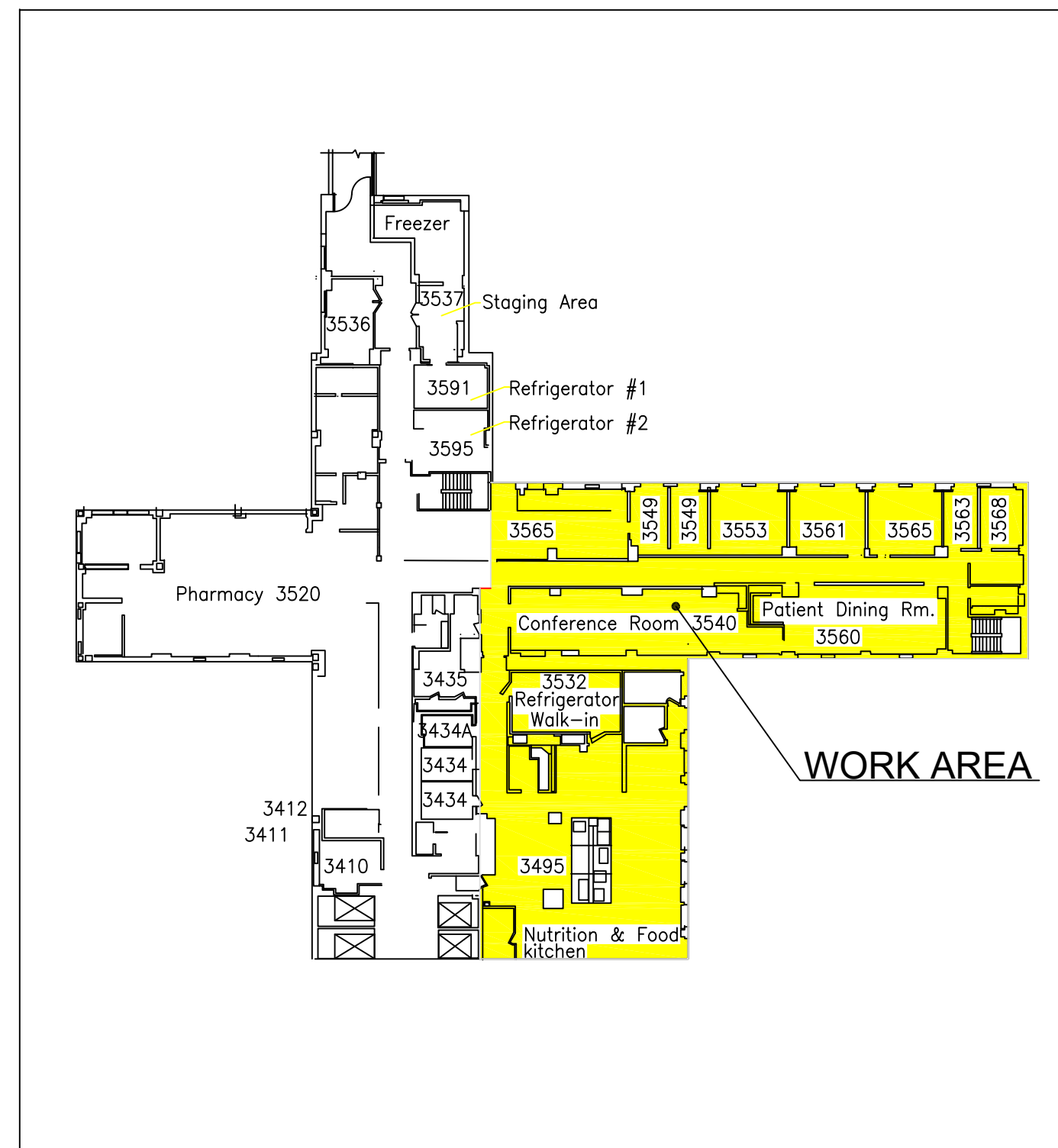


Department of
Veterans Affairs

AHU-AC21 REPLACEMENT SERVING NUTRITION FOOD AND SERVICE
DESIGN, 3RD FLOOR DAMEN BUILDING 1-B
JESSE BROWN VETERANS AFFAIRS MEDICAL CENTER
820 SOUTH DAMEN AVENUE
CHICAGO, IL 60612
PROJECT 537-26-200
BID DOCUMENTS



FACILITY MAP
NTS



WORK AREA - BLDG 1-B, 3RD FLOOR
NTS

INDEX OF DRAWINGS:

G-1	TITLE SHEET AND LOCATION PLAN
G-2	GENERAL NOTES AND CONDITIONS
MD-1	DEMOLITION OF FAN COIL AND FIN TUBE RADIATION SYSTEMS
MD-2	MECHANICAL DEMOLITION BUILDING 1-B ROOF
MD-3	PHOTOGRAPHS - MECHANICAL DEMOLITION
M-1	PIPING NOTES AND MECHANICAL SCHEDULE I
M-2	MECHANICAL SCHEDULE II
M-3	MECHANICAL SCHEDULE III
M-4	MECHANICAL DETAILS I
M-5	MECHANICAL DETAILS II
M-6	HOT WATER PIPING DIAGRAMS
M-7	GENERAL LAYOUT FOR NEW AHU AC-21
M-8	REVISIONS TO HYDRONIC PIPING
TC-1	TEMPERATURE CONTROL DIAGRAMS I
TC-2	TEMPERATURE CONTROL DIAGRAMS II
V-1	VENTILATION FLOW RATES

APPLICABLE CODES AND STANDARDS

ALL WORK SHALL COMPLY TO:

1. ALL VA POLICIES
2. VA HVAC DESIGN MANUAL, March 1, 2024
3. ASHRAE 170-2017 VENTILATION OF HEALTH CARE FACILITIES
4. VA CONSTRUCTION SAFETY
5. VA INFECTION CONTROL POLICIES
6. VA CODES & STANDARDS, VA DIRECTIVES AND EXECUTIVE ORDERS FOUND AT:
<http://www.cfm.va.gov/til/>
7. 2023 INTERNATIONAL MECHANICAL CODE
8. 2023 INTERNATIONAL ENERGY CODE
9. 2024 NFPA-70, NATIONAL ELECTRIC CODE

NOTE: CONTRACTOR SHALL OBTAIN AND VERIFY ALL DIMENSIONS AND CONDITIONS AT JOB SITE AND BE FULLY RESPONSIBLE FOR SAME. BRIDGING DOCUMENTS, DRAWINGS AND NARRATIVES ARE PROVIDED FOR DESIGN INTENT. THE DESIGN-BUILD ENTITY IS RESPONSIBLE FOR THE COMPLETE DESIGN OF A PROJECT THAT ADHERES TO ALL SCOPE OF WORK REQUIREMENTS, CODES, STATE AND FEDERAL REGULATIONS AND GUIDELINES.

FULLY SPRINKLERED

Revisions:		Date		Revisions Continued:		Date		VA STATION VA HEALTH CARE EXCELLENCE in the 21st Century Jesse Brown VA Medical Center 820 S. Damen Ave. Chicago, IL 60612		Office of Construction and Facilities Management U.S. Department of Veterans Affairs		Drawing Title TITLE SHEET AND LOCATION PLAN Approved Project Director		Phase BID DOCUMENTS FULLY SPRINKLERED		Project Title AHU-AC21 REPLACEMENT SERVING NUTRITION FOOD AND SERVICE Location JESSE BROWN VAMC Date: October 30, 2025		Project Number 537-26-200 Building Number 1-B Drawing Number G-1 Dwg. 1 of 16		Office of Planning and Construction Department of Veterans Affairs	
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1. CONTRACTOR SHALL PROVIDE ALL LABOR, MATERIALS, EQUIPMENT, AND SERVICES NECESSARY TO COMPLETE THE WORK INDICATED IN THE CONTRACTED BID DOCUMENTS, INCLUDING THE STATEMENT OF WORK (SOW). WORKMANSHIP SHALL BE COMPLETE IN EVERY ASPECT, TESTED, APPROVED, SATISFACTORY TO JESSE BROWN VA MEDICAL CENTER (JB VAMC), AND IN ACCORDANCE WITH VA CODES AND STANDARDS, VHA DIRECTIVES, FEDERAL LAWS, AND STATE AND LOCAL REGULATIONS.
2. THE DRAWINGS INDICATE DIAGRAMMATICALLY THE EXTENT AND LOCATION OF THE WORK INCLUDED. WORK INDICATED, BUT HAVING MINOR DETAILS OBVIOUSLY OMITTED, SHALL BE PROVIDED, INCLUDING THESE DETAILS, WITHOUT EXTRA COST.
3. CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY APPROVALS AND PERMITS REQUIRED FROM JB VAMC BEFORE STARTING ANY WORK.
4. THE WORK ON THESE DRAWINGS IS INTENDED TO CONVEY THE SCOPE OF WORK AND INDICATE THE GENERAL ARRANGEMENT OF EQUIPMENT AND FINAL LAYOUT.
5. THE CLAUSE ENTITLED "SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION " IN FAR 52.236-21 (APRIL 2019) IS SUPPLEMENTED AS FOLLOWS:
 - (a) THE CONTRACTING OFFICER'S INTERPRETATION OF THE DRAWINGS AND SPECIFICATIONS WILL BE FINAL, SUBJECT TO THE DISPUTES CLAUSE.
 - (b) THE CONTRACTOR SHALL:
 - a. CHECK ALL DRAWINGS AND SPECIFICATIONS FURNISHED IMMEDIATELY UPON RECEIPT.
 - b. COMPARE ALL DRAWINGS AND SPECIFICATIONS AND VERIFY THE FIGURES BEFORE LAYING OUT THE WORK.
 - c. PROMPTLY NOTIFY THE CONTRACTING OFFICER OF ANY DISCREPANCIES.
 - d. BE RESPONSIBLE FOR ANY ERRORS THAT MAY HAVE BEEN AVOIDED BY COMPLYING WITH PARAGRAPH (b); AND
 - e. REPRODUCE AND PRINT CONTRACT DRAWINGS AND SPECIFICATIONS AS NEEDED.
 - (a) IN GENERAL:
 - a. DRAWINGS OF GREATER DETAIL SHALL GOVERN DRAWINGS OF LESSER DETAIL UNLESS SPECIFICALLY NOTED OTHERWISE; AND
 - b. FIGURES AND NUMERICAL QUANTITIES NOTED ON DRAWINGS GOVERN OVER SCALE MEASUREMENTS.
 - (a) OMISSIONS FROM THE DRAWINGS OR SPECIFICATIONS OR THE MISDESCRIPTION OF DETAILS OF WORK THAT ARE MANIFESTLY NECESSARY TO CARRY OUT THE INTENT OF THE DRAWINGS AND SPECIFICATIONS, OR THAT ARE CUSTOMARILY PERFORMED, SHALL NOT RELIEVE THE CONTRACTOR FROM PERFORMING SUCH OMITTED OR MISDESCRIBED DETAILS OF THE WORK. THE CONTRACTOR SHALL PERFORM SUCH DETAILS AS IF FULLY AND CORRECTLY SET FORTH AND DESCRIBED IN THE DRAWINGS AND SPECIFICATIONS.
6. CONTRACTOR SHALL FIELD VERIFY ALL DIMENSIONS AND LOCATION OF NEW CONSTRUCTION WITH EXISTING WALLS, FLOORS, CEILINGS, COLUMNS, MECHANICAL AND ELECTRICAL EQUIPMENT, AND EXISTING CONDITIONS.
7. CONTRACTORS ARE RESPONSIBLE FOR CHECKING FIELD CONDITIONS PRIOR TO THE COMMENCEMENT OF WORK. CONTRACTORS SHALL REPORT ANY PROBLEMS/CONFLICTS TO THE VA COR WITHIN TWO (2) DAYS OF DISCOVERY. CONTRACTOR IS RESPONSIBLE FOR ANY CHANGES RESULTING FROM CONDITIONS ARISING IN THE FIELD WHICH ARE NOT BROUGHT TO THE ATTENTION OF THE VA COR AND MADE BY THE CONTRACTOR AT NO ADDITIONAL COST TO VA JESSE BROWN MEDICAL CENTER .
8. FINAL LOCATION OF HVAC COMPONENTS AND EQUIPMENT SHALL BE APPROVED BY THE VA COR.
9. CONTRACTORS SHALL WARRANT ALL WORK TO BE PERFORMED AND ALL MATERIALS TO BE FURNISHED UNDER THIS CONTRACT AGAINST DEFECTS IN WORKMANSHIP IN ACCORDANCE WITH VA'S STANDARD DOCUMENTS FOR CONSTRUCTION.
10. IN ACCORDANCE WITH GENERAL CONSTRUCTION PRACTICES, CONTRACTORS SHALL BE SOLELY RESPONSIBLE FOR WORKING CONDITIONS AT THE JOB SITE.
11. CONTRACTORS SHALL ARRANGE FOR THE PREMISES TO BE MAINTAINED IN A CLEAN AND ORDERLY MANNER THROUGHOUT THE COURSE OF THE CONSTRUCTION. DO NOT BLOCK ENTRANCES, EXITS, LOBBIES, ETC. LEAVE AREAS CLEAN AND FREE OF DEBRIS AT THE END OF EACH DAY.
12. CONTRACTORS SHALL BE RESPONSIBLE FOR REMOVING ALL LEFT-OVER MATERIALS, DEBRIS, AND EQUIPMENT INVOLVED IN HIS OPERATIONS AT THE CONCLUSION OF THE PROJECT. ALL PROJECT AREAS AND SURROUNDING AREAS AFFECTED BY THE CONSTRUCTION SHALL BE LEFT CLEAN AND IN GOOD CONDITION.
13. CONTRACTORS SHALL TAKE PRECAUTIONARY MEASURES TO ENSURE THAT ALL PROPERTIES, EQUIPMENT, AND DEVICES AT THE PROJECT SITE ARE PROTECTED DURING THE CONSTRUCTION PERIOD. ANY DAMAGE TO EXISTING CONSTRUCTION, EQUIPMENT AND DEVICES SHALL BE REPAIRED AND RESTORED TO A CONDITION EQUAL TO THAT EXISTING AT THE COMMENCEMENT OF WORK.
14. ALL FINISH WORK THAT IS DAMAGED OR STAINED SHALL BE REPAIRED OR REPLACED IF NECESSARY AT NO ADDITIONAL EXPENSE TO JB VAMC.
15. CONTRACTORS SHALL IMMEDIATELY REPORT ANY ENCOUNTERED UNKNOWN OR SUSPECTING HAZARDOUS MATERIAL TO THE VA COR FOR TESTING.
16. ALL PIPING TO NEW EQUIPMENT SHALL BE THOROUGHLY FLUSHED BEFORE CONNECTING TO NEW EQUIPMENT.

A/E	ARCHITECT / ENGINEER
	MODULATING
ACD-TP	AUTOMATIC CONTROL DAMPER, TWO POSITION
ACU	AIR CONDITIONING UNIT
AD	ACCESS DOOR
AF	AFTER FILTER
AFF	ABOVE FINISHED FLOOR
AFMS	AIR FLOW MEASURING STATION
AHU	AIR-HANDLING UNIT
AMP	AMPERE
AP	ACCESS PANEL
APD	AIR PRESSURE DROP
AS	AIR SEPARATOR
ASHRAE	AMERICAN SOCIETY OF HEATING REFRIGERATION AIR CONDITIONING ENGINEERS
ASME	AMERICAN SOCIETY OF MECHANICAL ENGINEERS
BHP	BRAKE HORSEPOWER
BTU	BRITISH THERMAL UNIT
BTUH	BRITISH THERMAL UNIT PER HOUR
CC	COOLING COIL
CCD	COOLING COIL CONDENSATE DRAIN
CD	CEILING DIFFUSER
CFM	CUBIC FEET PER MINUTE
CFT	CUBIC FEET
CFP	CHEMICAL FEED PUMP
CG	CEILING GRILLE
CHW	CHILLER WATER
CHR	CHILLED WATER RETURN
CHS	CHILLED WATER SUPPLY
CM	CARBON MONOXIDE
COMP	COMPRESSOR UNIT
CU	CONDENSING UNIT
CWCC	CHILLED WATER COOLING COIL
CWP	CONDENSER WATER PUMP
CWR	CONDENSER WATER RETURN (TO COOLING TOWER)
CWS	CONDENSER WATER SUPPLY (FROM COOLING TOWER)
D	DAMPER - AUTOMATIC
Db	DRY-BULB TEMPERATURE
DDC	DIRECT DIGITAL CONTROLS
DEG	DEGREE
DF	DIFFUSER
DHW	DOMESTIC HOT WATER
DHWR	DOMESTIC HOT WATER RETURN
DIA	DIAMETER
DPS	DIFFERENTIAL PRESSURE SENSOR
DX	DIRECT EXPANSION

-----S-60-----	HIGH PRESSURE STEAM (60 PSIG AND ABOVE)
-----CR-60-----	HIGH PRESSURE STEAM CONDENSATE RETURN
-----S-30-----	MEDIUM PRESSURE STEAM (16 PSIG THRU 59 PSIG)
-----CR-30-----	MEDIUM PRESSURE STEAM CONDENSATE RETURN
-----S-15-----	LOW PRESSURE STEAM (15 PSIG AND BELOW)
-----CR-15-----	LOW PRESSURE STEAM CONDENSATE RETURN
-----PC-----	CONDENSATE PUMP DISCHARGE
-----HWS-----	HOT WATER HEATING SUPPLY
-----HWR-----	HOT WATER HEATING RETURN
-----GHS-----	GLYCOL-WATER HEATING SUPPLY
-----GHR-----	GLYCOL-WATER HEATING RETURN
-----CHS-----	CHILLED WATER SUPPLY
-----CHR-----	CHILLED WATER RETURN
-----GCS-----	CHILLED GLYCOL-WATER SUPPLY
-----GCR-----	CHILLED GLYCOL-WATER RETURN
-----MW-----	MAKE-UP WATER
-----D-----	DRAIN LINE
-----V-----	VENT LINE
-----GRS-----	GLYCOL-WATER RUN AROUND SUPPLY
-----GRR-----	GLYCOL-WATER RUN AROUND RETURN
-----X-----	EXISTING PIPE TO BE REMOVED

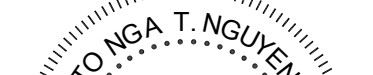
	CHECK VALVE
	WYE STRAINER (WITH BALL VALVE)
	FLEXIBLE CONNECTION
	ANGLE GLOBE VALVE
	BUTTERFLY VALVE
	BALL VALVE
	MODULATING CONTROL VALVE
	PRESSURE SAFETY VALVE
	MODULATING CONTROL BUTTERFLY VALVE
	WATER BALANCE DEVICE
	CIRCUIT SETTER VALVE

	DUCT INSIDE DIMENSIONS: WIDTH x DEPTH
	EXISTING DUCT TO REMAIN
	EXISTING DUCT TO BE REMOVED
	UNIT HEATER (HORIZONTAL)

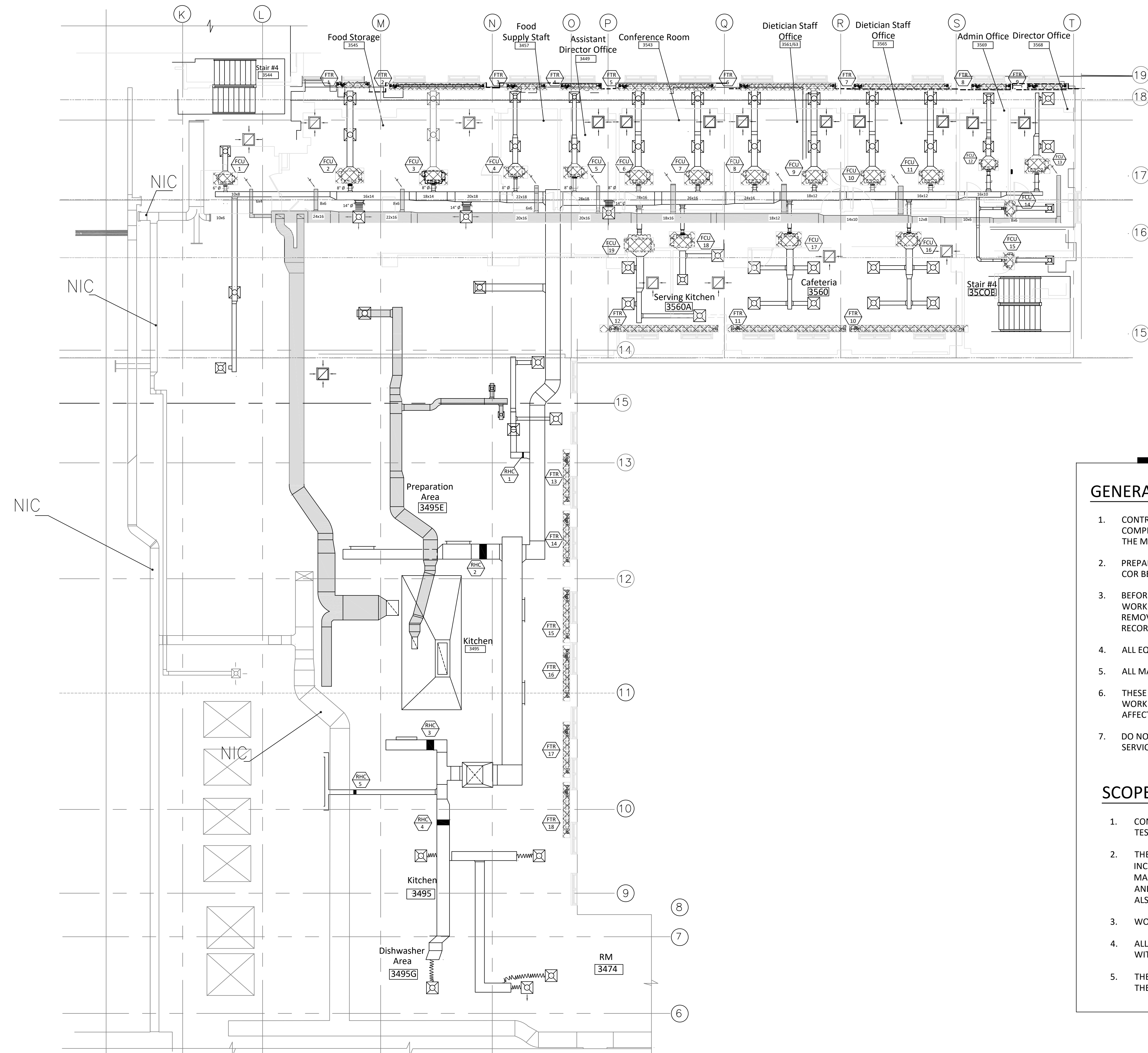
 ← DETAIL NUMBER
 ← DRAWING NUMBER WHERE DRAWN

 ← SECTION LETTER
 ← DRAWING NUMBER WHERE SHOWN

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				VA STATION  Defining EXCELLENCE in the 21st Century Jesse Brown VA Medical Center 820 S. Damen Ave. Chicago, IL 60612		 A. T. NGUYEN 062-050406 LICENSED PROFESSIONAL ENGINEER OF THE STATE OF ILLINOIS  EXPIRES NOV. 30, 2025		Office of Construction and Facilities Management  U.S. Department of Veterans Affairs		Drawing Title GENERAL NOTES AND CONDITIONS		Phase BID DOCUMENTS		Project Title AHU-AC21 REPLACEMENT SERVING NUTRITION FOOD AND SERVICE		Project Number 537-26-200		Office of Planning and Construction  Department of Veterans Affairs	
										Approved Project Director		FULLY SPRINKLERED		Location JESSE BROWN VAMC		Building Number 1-B			
Revisions:		Date		Revisions Continued:		Date						Date: October 30, 2025		Checked: D. Moreau		Prepared: A. Nguyen		Drawing Number G-2 Dwg. 2 of 16	

three inches = one foot
one and one half inches = one foot
one inch = one foot
three quarters inch = one foot
one half inch = one foot
three eighths inch = one foot
one quarter inch = one foot
one eighth inch = one foot



DEMOLITION WORK - BLDG 1-B, 3RD FLOOR
SCALE: 1/8" = 1'-0"

SYMBOL & ABBREVIATION LIST

- CW - COLD WATER PIPING
- HW - HOT WATER PIPING
- HWR - HOT WATER RETURN PIPING
- BALL VALVE
- GATE VALVE
- CHECK VALVE
- UNION
- POINT OF CONNECTION BETWEEN NEW & EXISTING
- LIMIT OF DEMOLITION
- A.P. ACCESS PANEL
- FIN TUBE RADIATORS
- FAN COIL UNIT TO BE REPLACED
- EXISTING SUPPLY DUCTWORK AND DIFFUSERS TO REMAIN
- EXISTING RETURN DUCTWORK AND DIFFUSERS TO REMAIN
- NIC NOT INCLUDED IN CONTRACT

GENERAL DEMOLITION NOTES

- CONTRACTOR SHALL REVIEW THE DRAWINGS AND SPECIFICATIONS, IN PARTICULAR SECTION 02 41 00 DEMOLITION, AND LAYOUT HIS WORK IN A COMPATIBLE AND COMPLEMENTARY MANNER, REMOVE ALL EQUIPMENT, SUPPORT, CONTROL ACCESSORIES, ETC, AND MECHANICAL ITEMS MADE OBSOLETE BY THESE ALTERATIONS AS SHOWN IN THE MECHANICAL DRAWINGS.
- PREPARE A DEMOLITION PLAN AND SUBMIT PROPOSED DEMOLITION AND REMOVAL PROCEDURES FOR APPROVAL BEFORE WORK IS STARTED. PLAN SHALL BE APPROVED BY THE COR BEFORE THE START OF WORK. DO NOT BEGIN DEMOLITION UNTIL AUTHORIZATION IS RECEIVED.
- BEFORE THE START OF THE DEMOLITION, THE CONTRACTOR SHALL SURVEY THE SITE AND EXAMINE THE DRAWINGS AND SPECIFICATIONS TO DETERMINE THE EXTENT OF THE WORK. RECORD EXISTING CONDITIONS IN THE PRESENCE OF THE COR. IDENTIFY CONDITIONS OF STRUCTURES AND OTHER FACILITIES ADJACENT TO THE AREAS OF ALTERATIONS OR REMOVAL. IT IS THE CONTRACTOR'S RESPONSIBILITY TO VERIFY AND DOCUMENT ALL WHICH WILL BE REQUIRED DURING THE COURSE OF THE WORK, AND TO NOTE THESE ON THE RECORD DOCUMENTS.
- ALL EQUIPMENTS AND DEVICES SHALL REMAIN UNLESS SPECIFICALLY NOTED FOR REMOVAL.
- ALL MATERIALS AND EQUIPMENT REMOVED AND NOT REUSED OR SALVAGED SHALL BECOME THE PROPERTY OF THE CONTRACTOR AND SHALL BE REMOVED FROM SITE.
- THESE DRAWINGS ARE COMPILED BY THE VA FROM VA'S RECORD DRAWINGS AND LIMITED FIELD VERIFICATION OF EXISTING CONDITIONS FOR THE PURPOSE OF INDICATING THE WORK REQUIRED AND ARE BELIEVED TO BE CORRECT, NOTWITHSTANDING, THE CONTRACTOR SHALL VERIFY EQUIPMENT LOCATIONS, DIMENSIONS, AND ALL FIELD CONDITIONS AFFECTING HIS WORK.
- DO NOT INTERRUPT EXISTING UTILITIES SERVICING OCCUPIED OR USED FACILITIES, EXCEPT WHEN APPROVED IN WRITING BY THE COR AND THEN ONLY AFTER TEMPORARY UTILITY SERVICES HAVE BEEN APPROVED AND PROVIDED.

SCOPE OF WORK NOTES

- CONTRACTOR SHALL INSPECT THE INTEGRITY OF THE SUPPLY AND EXHAUST DUCT WORK AND COMPLETE ALL REQUIRED REPAIRS AND CORRECTIONS, INCLUDING PRESSURE TESTING. THE EXISTING DUCTWORK IS SERVED BY SUPPLY AIR HANDLER AC-1, KITCHEN EXHAUST FANS E-2, AND RV-16.
- THE CONTRACTOR SHALL REPLACED THE 19 TWO-PIPE FAN COIL UNITS SHOWN ON THIS DRAWING. THE NEW FAN COIL UNITS SHALL BE OF THE 2-PIPE TYPE AND SHALL INCLUDE THE REQUIRED CONTROL VALVES AND DDC CONTROLLER/SENSORS. THE NEW DDC CONTROL SYSTEM I IN THIS PROJECT SHALL BE INCORPORATED INTO THE BUILDING MANAGEMENT SYSTEM (BMS) AT JESSE BROWN VAMC ECC. THE PROJECT ALSO CALLS FOR THE REPLACEMENT OF FIN TUBE RADIATOR SYSTEM, INCLUDING CONTROL VALVES AND THERMOSTATIC TRAPS, FIVE KITCHEN REHEAT COILS, AND A STEAM PRESSURE REDUCING STATION (20 TO 5 PSIG) BY THE POT WASHING AREA (Rm 3495A). WORK ALSO CALLS FOR THE REPLACEMENT OF THE KITCHEN EXHAUST FANS E-2 AND RV-16, AND THE SUPPLY AIR HANDLER AC-21, LOCATED ON THE ROOF OF BUILDING 1-B.
- WORK ALSO INCLUDES THE DEMOLITION OF AHU AC-35 AND ASSOCIATED DUCTWORK AND EQUIPMENT IDENTIFIED ON DWG MD-3, PHOTOGRAPH # 9.
- ALL EXISTING THERMOSTATS SHALL BE REMOVED AND REPLACED AT THEIR CURRENT LOCATIONS. ALL THERMOSTAT COMMUNICATION WIRING TO REMAIN AND BE REUSE D WITH THE NEW THERMOSTATS.
- THE NEW HVAC SYSTEMS MUST BE TESTED AND BALANCE BEFORE THE START OF THE DEMOLITION TO IDENTIFY AREAS OF CORRECTIONS (ESTABLISH A BASE LINE), AND AFTER THE INSTALLATION IS COMPLETED TO VERIFY COMPLIANCE WITH THE DESIGN DOCUMENTS.

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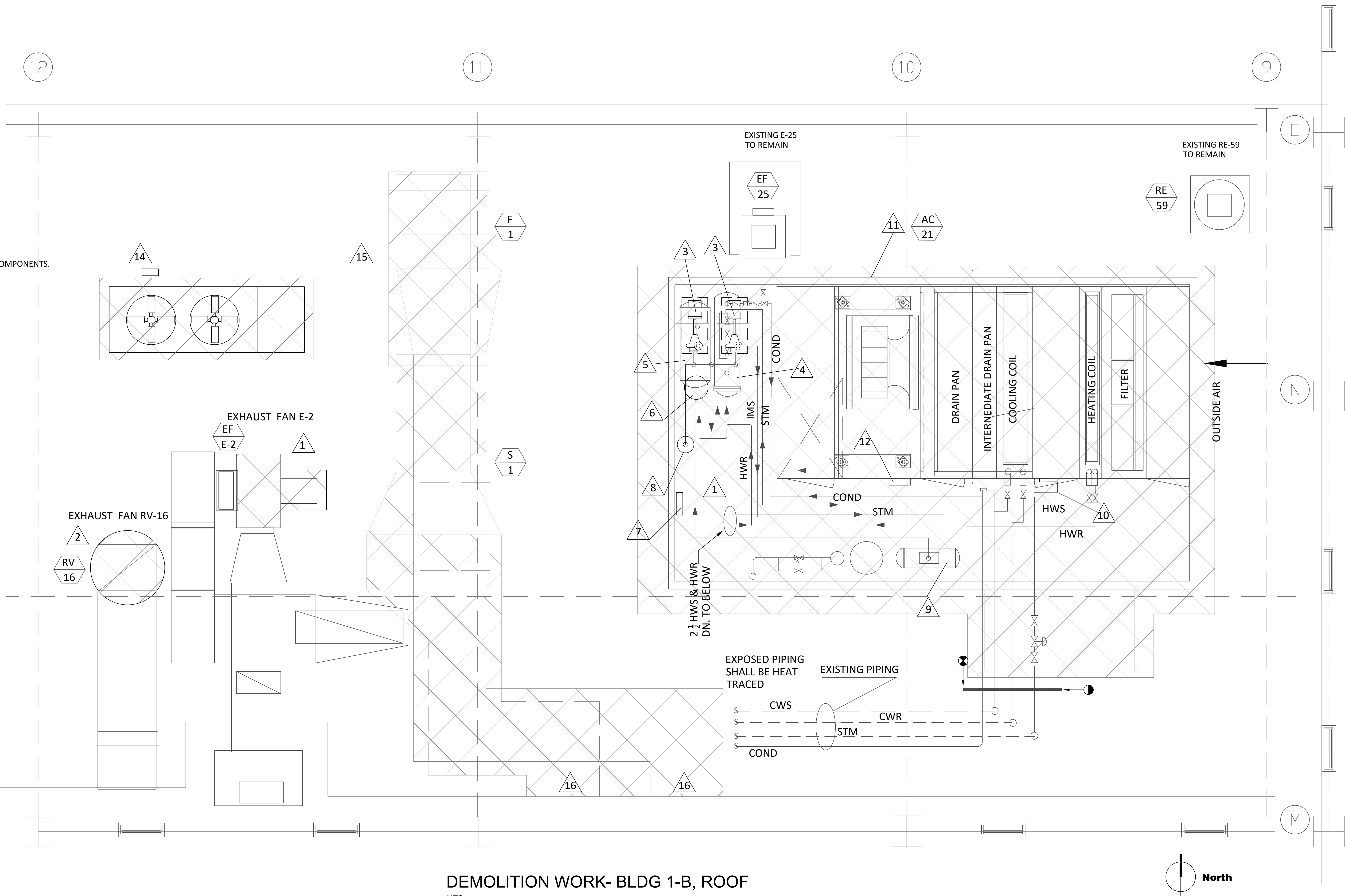
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EQUIPMENT ASSOCIATED WITH AC-21 PROJECT TO BE REMOVED AND REPLACED

1. EXHAUST FAN E-2, DUCTWORK TO REMAIN.
2. EXHAUST FAN RV-16, DUCTWORK TO REMAIN.
3. HOT WATER PUMPS P-1 & P-2 AND HOUSEKEEPING PADS.
4. HEAT EXCHANGER HX-1.
5. AIR SEPARATOR AS-1.
6. EXPANSION TANK EX-1.
7. BUILDING AUTOMATION CONTROL BAC.
8. CHEMICAL POT CPT-1.
9. PRESSURE FILL SYSTEM PFS-1.
10. ELECTRIC UNIT HEATER EUH-1.
11. AIR HANDLER AC-21.
12. FAN VARIABLE FREQUENCY DRIVE.
13. ELECTRICAL PANEL & PUMP STARTERS.
14. CONDENSER & ASSOCIATED ELECTRICAL COMPONENTS.
15. AIR HANDLER/DX COIL & SUPPLY/RETURN DUCTWORK & ELECTRICAL COMPONENTS.
16. AFTER REMOVAL OF DUCTWORK, CONTRACTOR SHALL INSTALL AN INSULATED ALUMINUM WALL COVER.

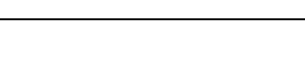
GENERAL DEMOLITION NOTES

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2. PREPARE A DEMOLITION PLAN AND SUBMIT PROPOSED DEMOLITION AND REMOVAL PROCEDURES FOR APPROVAL BEFORE WORK IS STARTED. PLAN SHALL BE APPROVED BY THE COR BEFORE THE START OF WORK. DO NOT BEGIN DEMOLITION UNTIL AUTHORIZATION IS RECEIVED.
3. BEFORE THE START OF THE DEMOLITION, THE CONTRACTOR SHALL SURVEY THE SITE AND EXAMINE THE DRAWINGS AND SPECIFICATIONS TO DETERMINE THE EXTENT OF THE WORK. RECORD EXISTING CONDITIONS IN THE PRESENCE OF THE COR. IDENTIFY CONDITIONS OF STRUCTURES AND OTHER FACILITIES ADJACENT TO THE AREAS OF ALTERATIONS OR REMOVAL. IT IS THE RESPONSIBILITY OF THE CONTRACTOR'S RESPONSIBILITY TO TO VERIFY AND DOCUMENT ALL WHICH WILL BE REQUIRED DURING THE COURSE OF THE WORK, AND TO NOTE THESE OUTAGES ON THE RECORD DOCUMENTS.
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7. DO NOT INTERRUPT EXISTING UTILITIES SERVING OCCUPIED OR USED FACILITIES, EXCEPT WHEN APPROVED IN WRITING BY THE COR AND THEN ONLY AFTER TEMPORARY UTILITY SERVICES HAVE BEEN APPROVED AND PROVIDED. REMOVAL OF EQUIPMENT REQUIRES THE DISCONNECT OF ELECTRICAL POWER AND THE TERMINATION OF THE ELECTRICAL FEEDERS IN A WEATHER PROOF, WALL MOUNTED, JB BOX.
8. EQUIPMENT MUST BE ENERGIZED AT THE SERVICE PANELS BEFORE IT IS REMOVED. TERMINATE POWER LEADS AT THE BUILDING WALL IN WEATHER PROOF IN NEMA 3R ENCLOSURES.



DEMOLITION WORK- BLDG 1-B, ROOF
NTS

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				 Defining EXCELLENCE in the 21st Century						MECHANICAL DEMOLITION BUILDING 1-B ROOF		BID DOCUMENTS		AHU-AC21 REPLACEMENT SERVING NUTRITION FOOD AND SERVICE		537-26-200			
										Approved Project Director				Location JESSE BROWN VAMC		Building Number 1-B			
				Jesse Brown VA Medical Center 820 S. Damen Ave. Chicago, IL, 60612				 U.S. Department of Veterans Affairs				FULLY SPRINKLERED		Date: October 30, 2025		Checked: D. Moreau			
Revisions:				Date		Revisions Continued:		Date								MD-2		Dwg. 4 of 16	

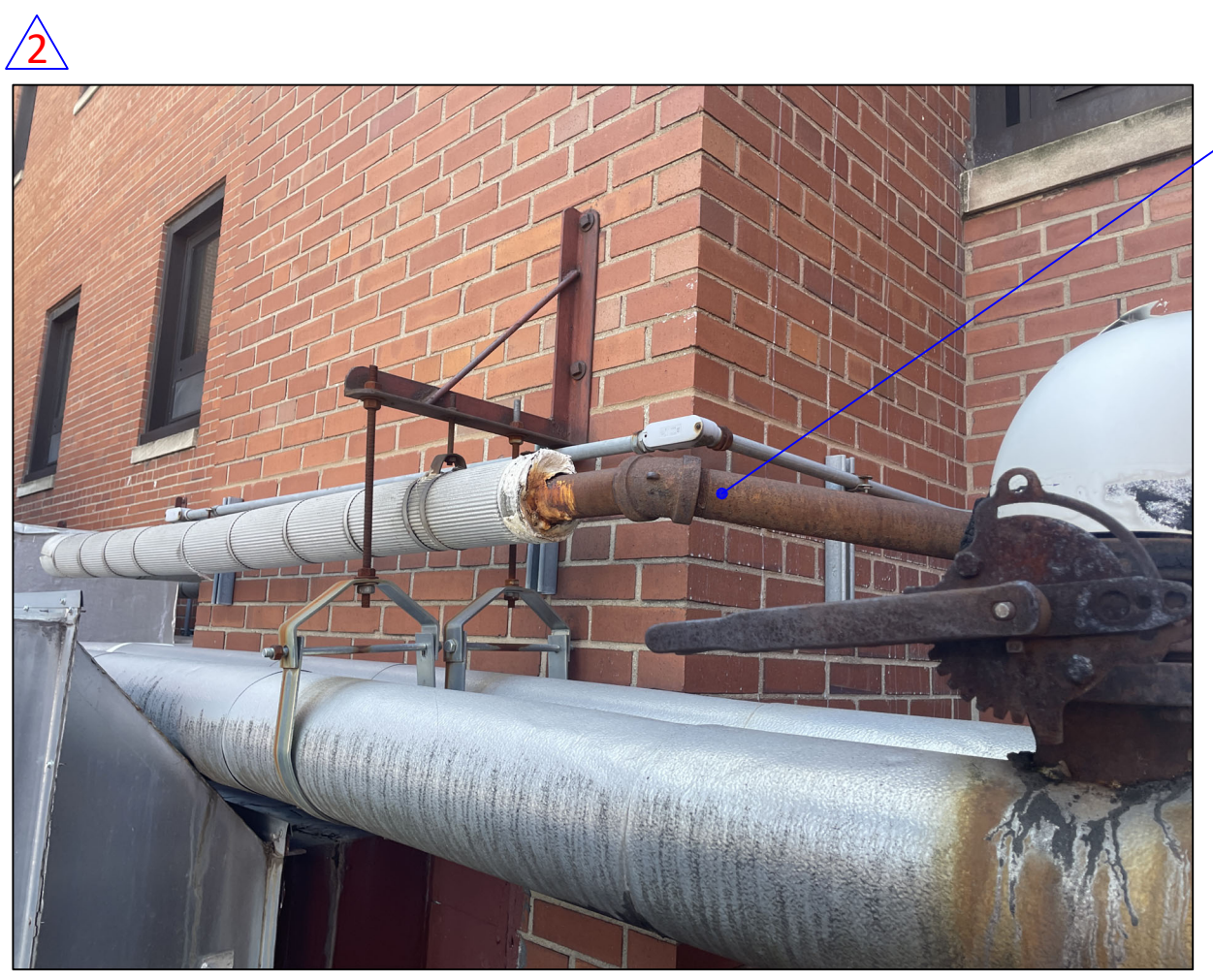
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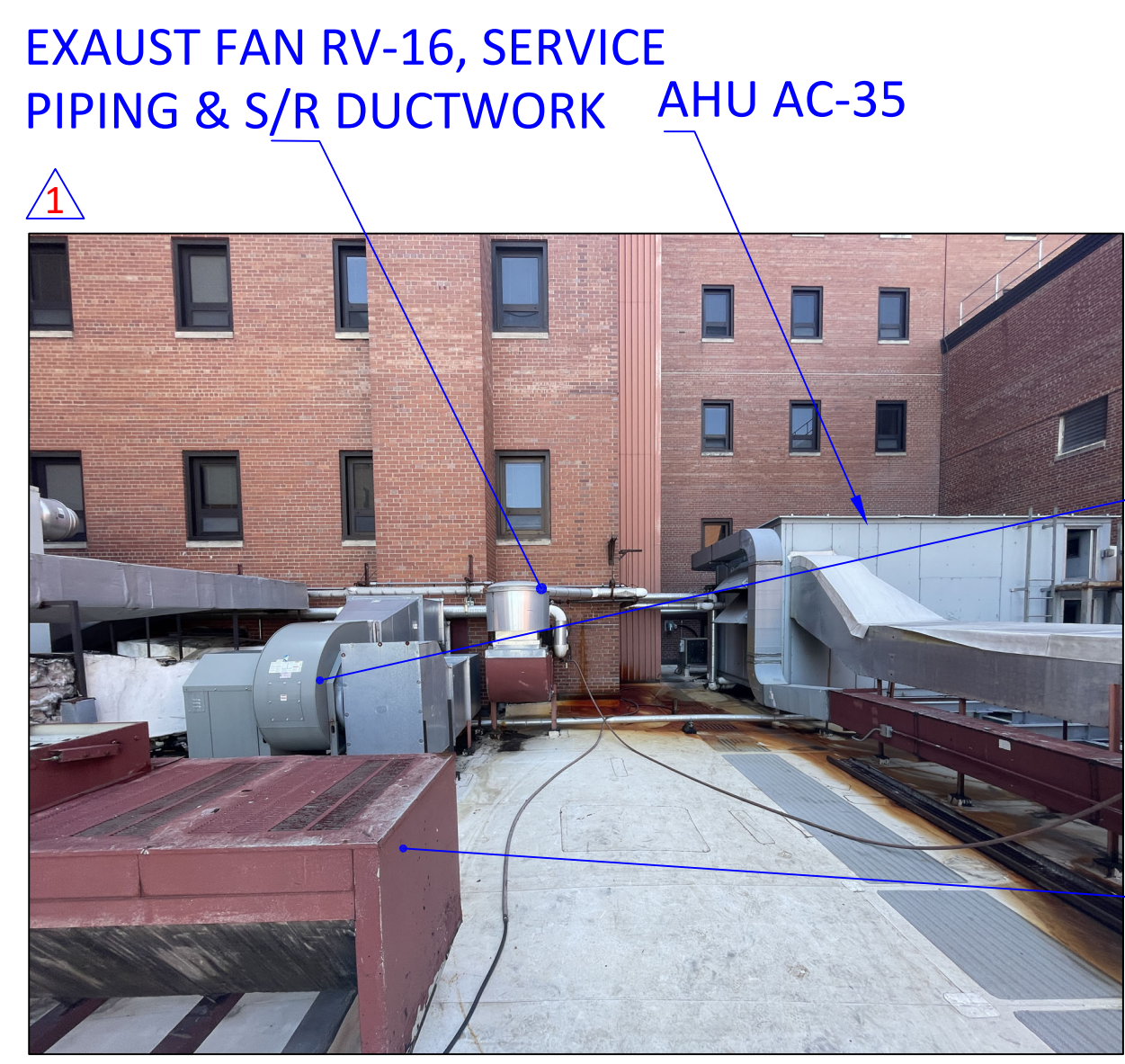
1. REMOVE AND DISPOSE OF EQUIPMENT IN PHOTOGRAPHS.
2. REMOVE DAMAGED INSULATION AND REINSULATE PIPING.
3. REPLACE FLEXIBLE CONNECTORS ON NEW FAN COIL UNITS.
4. TYPE OF OA LOUVER TO COOL EQUIPMENT NEW AHU AC-21.
5. EQUIPMENT TO REMAIN IN PLACE.
6. REMOVE NON-LOAD BEARING STEEL FRAME.



1 NORTH EAST VIEW OF ROOF



2 PIPING MISSING INSULATION



3 SOUTH VIEW OF ROOF



4 FAN COIL UNITS TO BE REPLACED



5 HOT WATER PUMPS P-1 & P-2 OPERATING INSIDE AHU AC21



6 EXAMPLE OF 6'x2" COOLING EQUIPMENT LOUVERS ON NEW AHU AC-21



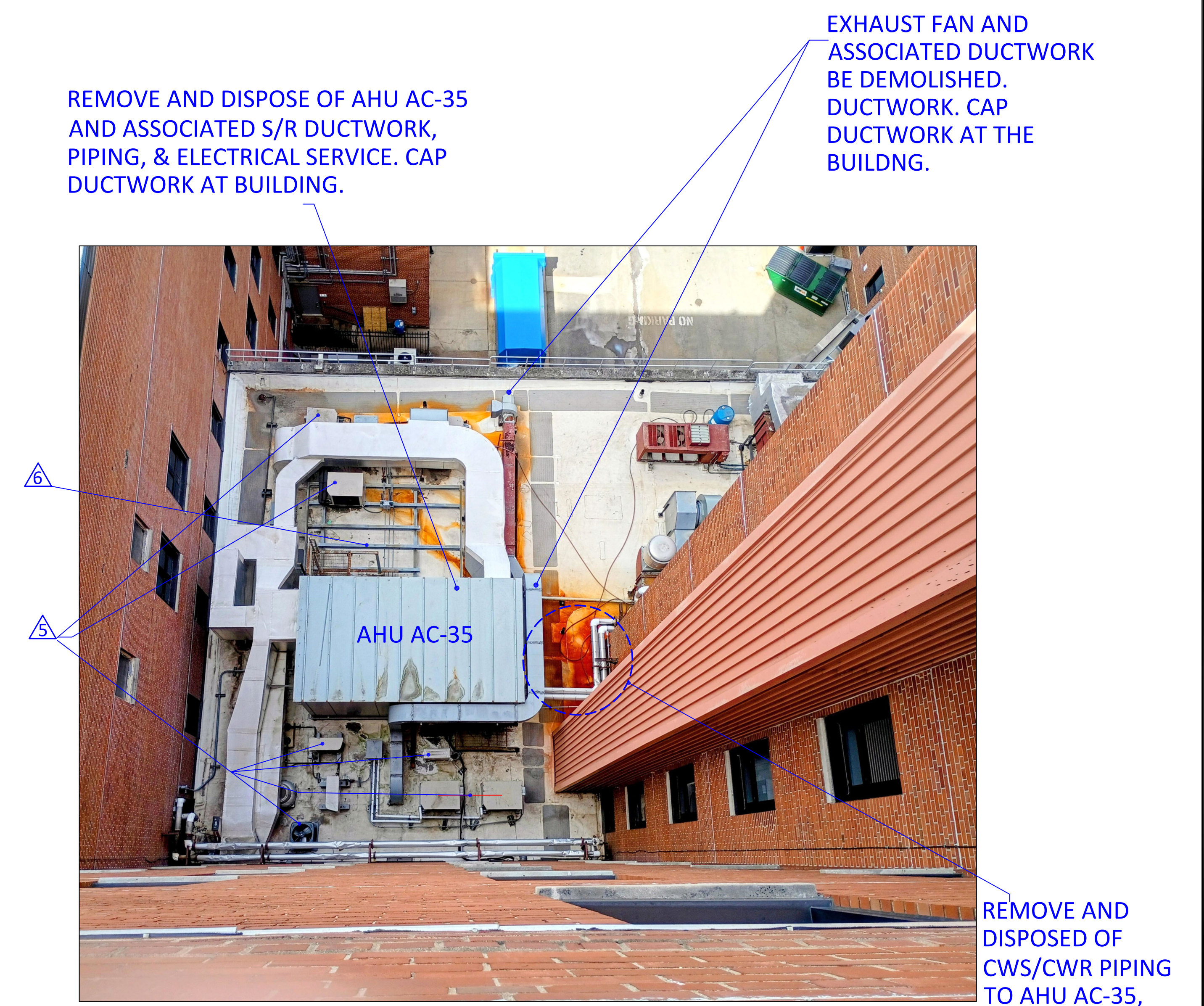
7 SUPPLY PIPING TO AHU AC21

CWS/CWR PIPING TO AHU AC-35 TO BE REMOVED, SEE PHOTO #9 ON THIS DWG.

REMOVE VALVES, INSTALL BLIND FLANGES, & REMOVE PIPING TO AHU AC-35.



8 CONDENSATE PIPING UNDER AHU AC-21 TO BE REUSED



9 WEST VIEW OF BUILDING B ROOF

NOTE: CONTRACTOR SHALL OBTAIN AND VERIFY ALL DIMENSIONS AND CONDITIONS AT JOB SITE AND BE FULLY RESPONSIBLE FOR SAME. BRIDGING DOCUMENTS, DRAWINGS AND NARRATIVES ARE PROVIDED FOR DESIGN INTENT. THE DESIGN-BUILD ENTITY IS RESPONSIBLE FOR THE COMPLETE DESIGN OF A PROJECT THAT ADHERES TO ALL SCOPE OF WORK REQUIREMENTS, CODES, STATE AND FEDERAL REGULATIONS AND GUIDELINES.

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16

FAN COIL UNIT SCHEDULE																		
MARK	AREA SERVED	DESIGN CFM	HOT WATER HEATING COIL (EAT:60°F)					CHILLED WTR COOLING COIL (EAT: 80/67°F)					ELECTRICAL REQUIREMENTS	FAN DATA			REMARKS	
			HTG CAPACITY TOTAL MBH	FLOW (GPM)	ΔP (FT.)	EWTT (°F)	ΔT (°F)	CLG CAPACITY TOTAL (MBH)	SENSIBLE (MBH)	FLOW (GPM)	ΔP (FT.)	EWTT (°F)		ΔT (°F)	WATTS	RPM		DRIVE
FCU-1	RM 3544	227	25.4			180		40	40	40	40	15	115/1/60			BELT		
FCU-2	RM 3445	550	15.8	4.9				5.0	5.0	5.0								
FCU-3	RM 3545	550	15.8	4.9				5.0	5.0	5.0								
FCU-4	RM 3547	378	43.5	3.6				3.6	3.6	3.6								
FCU-5	RM 3549	378	43.5	3.6				3.6	3.6	3.6								
FCU-6	RM 3553	512	62.6	4.9				5.0	5.0	5.0								
FCU-7	RM 3553	512	62.6	4.9				5.0	5.0	5.0								
FCU-8	RM 3561 RM 3563	512	62.6	4.9				5.0	5.0	5.0								
FCU-9	RM 3561 RM 3563	512	62.6	4.9				5.0	5.0	5.0								
FCU-10	RM 3565	512	62.6	4.9				5.0	5.0	5.0								
FCU-11	RM 3565	512	62.6	4.9				5.0	5.0	5.0								
FCU-12	RM 3569	371	43.5	2.1				3.6	3.6	3.6								
FCU-13	RM 3568	394	62.6	2.1				4.9	4.9	4.9								
FCU-14	RM 3566	214	25.4	5.3				2.1	2.1	2.1								
FCU-15	RM 3575	214	25.4	5.3				2.1	2.1	2.1								
FCU-16	RM 3560	613	68.2	5.3				5.3	5.3	5.3								
FCU-17	RM 3560	613	68.2	5.3				5.3	5.3	5.3								
FCU-18	RM 3560	613	68.2	5.3				5.3	5.3	5.3								
FCU-19	RM 3560A	736	81.1	6.1				6.3	6.3	6.3								
NOTES: 1. COOLING COIL SHALL BE SELECTED AT A MINIMUM OF 14°F TEMPERATURE RISE 2. HEATING COIL SHALL BE SELECTED AT A MINIMUM 20°F TEMPERATURE RISE																		

PUMP SCHEDULE										
MARK	SERVICE	PUMP DATA				ELEC. CHAR.			BASIS OF DESIGN	REMARKS
	LOCATION	SERVICE	TYPE	FLOW (GPM)	TDH (FT. of H ₂ O)	RPM	MOTOR HP	VOLT/φ/HZ		
P-1	ROOF (INSIDE AC-21)	HW	CENTRIF END SUCT.	395	105	1750	25	208/3/60	BELL & GOSSETT: 1510	
P-2	ROOF (INSIDE AC-21)	HW	CENTRIF END SUCT.	395	105	1750	25	208/3/60	BELL & GOSSETT: 1510	STANDBY

AIR SEPARATOR SCHEDULE				
MARK	SERVICE	FLOW RATE (GPM)	CONN. SIZE (IN)	REMARKS
	LOCATION			
AS-1	HW HTG SYSTEM AC-21	40	2-1/2	SEE NOTE 1

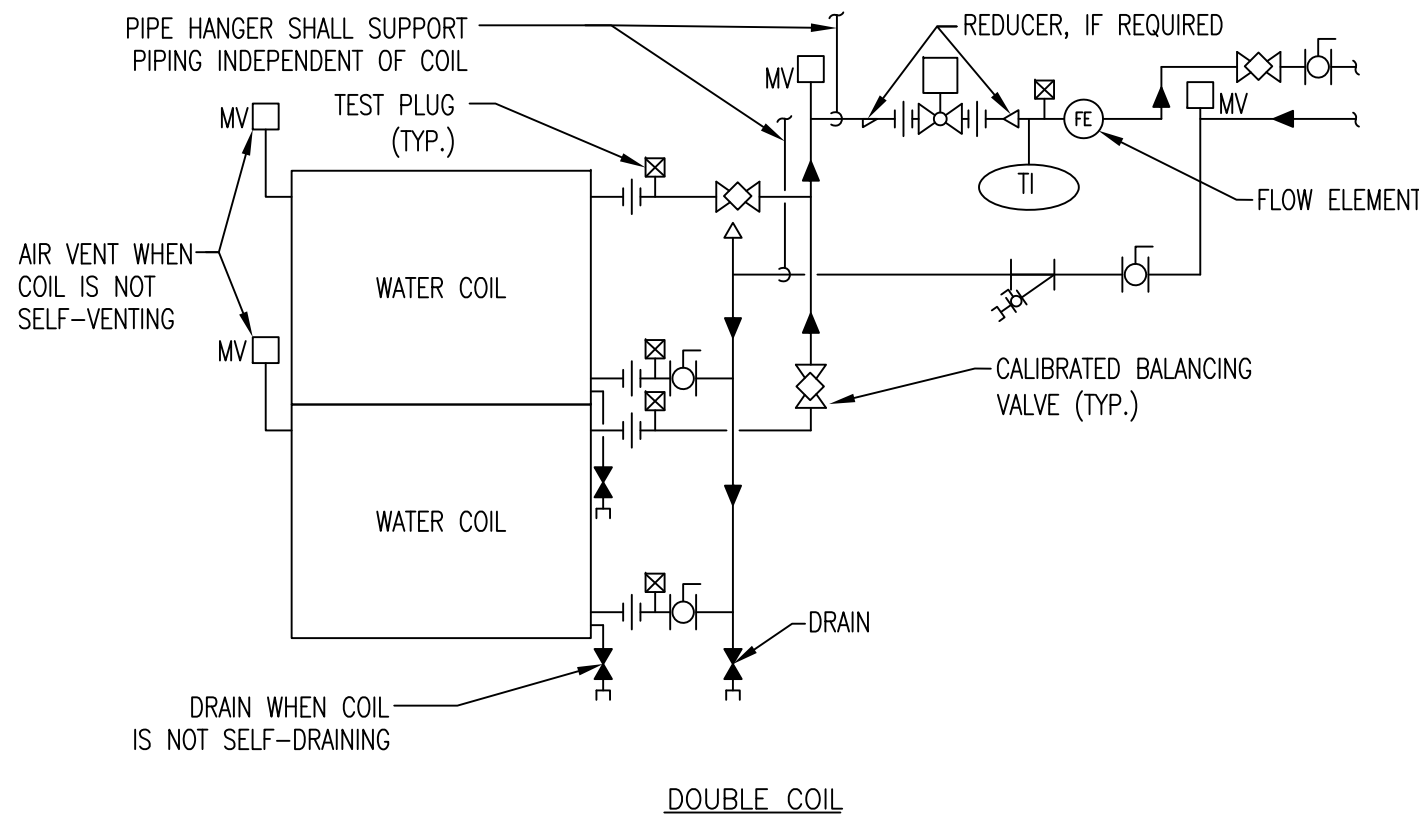
- NOTES:
1. INSTALLATION TO INCLUDE INTEGRAL STRAINER

NOTE: CONTRACTOR SHALL OBTAIN AND VERIFY ALL DIMENSIONS AND CONDITIONS AT JOB SITE AND BE FULLY RESPONSIBLE FOR SAME. BRIDGING DOCUMENTS, DRAWINGS AND NARRATIVES ARE PROVIDED FOR DESIGN INTENT. THE DESIGN-BUILD ENTITY IS RESPONSIBLE FOR THE COMPLETE DESIGN OF A PROJECT THAT ADHERES TO ALL SCOPE OF WORK REQUIREMENTS, CODES, STATE AND FEDERAL REGULATIONS AND GUIDELINES.

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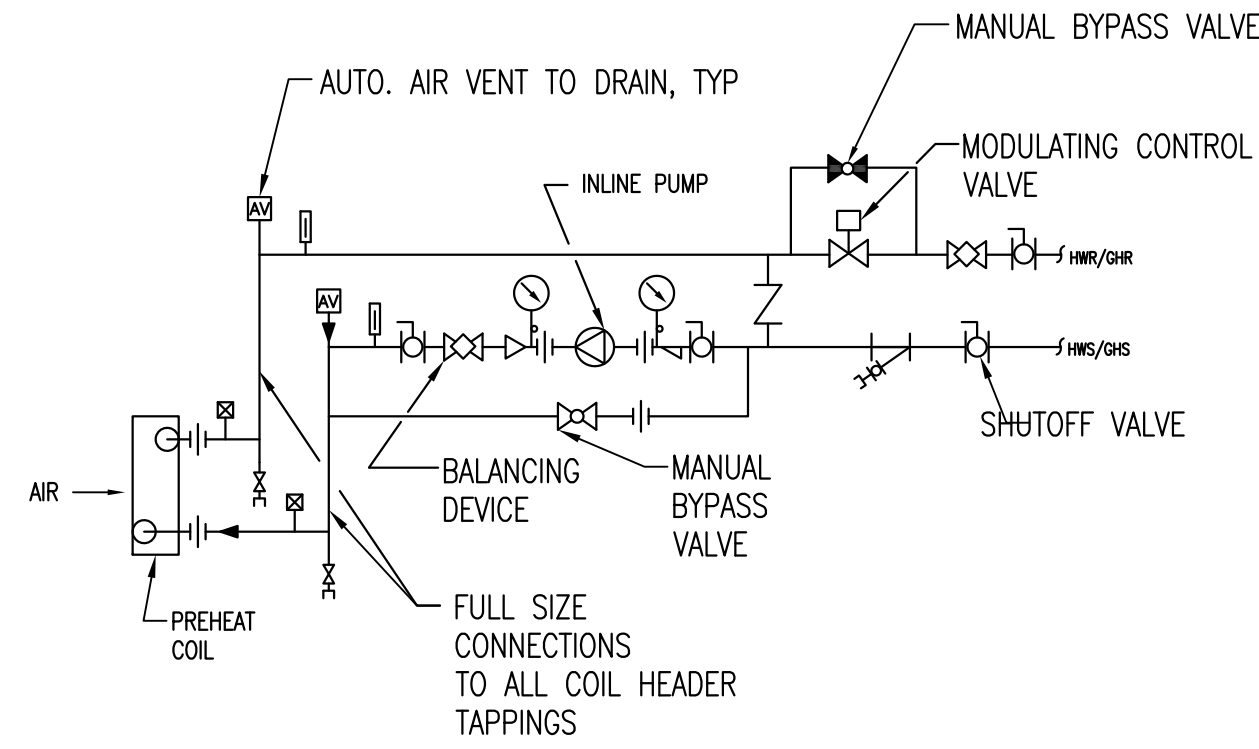
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one and one half inches = one foot
one inch = one foot
three quarters inch = one foot
one half inch = one foot
three eighths inch = one foot
one quarter inch = one foot
one eighth inch = one foot



- NOTES:
1. WHEN COIL IS INCLUDED IN CASING MOUNTED ON VIBRATION ISOLATORS THE FIRST 2 HANGERS FOR EACH PIPE SHALL BE SPRING & NEOPRENE TYPE. TYPE "H" FOR 100mm [4"] ϕ PIPE & SMALLER. TYPE "H-P" FOR 125mm [5"] ϕ PIPE & LARGER.
 2. PIPING SHALL BE INSTALLED IN SUCH MANNER THAT IT WILL NOT BLOCK THE SWING OR USE OF ACCESS DOORS OR PANELS; NEITHER SHALL IT BLOCK THE SERVICING OF FILTERS, VALES, OR EQUIPMENT.
 3. THE FLOW ELEMENT MAY BE INSTALLED IN THE SUPPLY PIPING IF THE REQUIRED MINIMUM UPSTREAM AND DOWNSTREAM DIMENSIONS CANNOT BE OBTAINED IN THE RETURN PIPING.

1 WATER COILS - PIPING CONNECTIONS

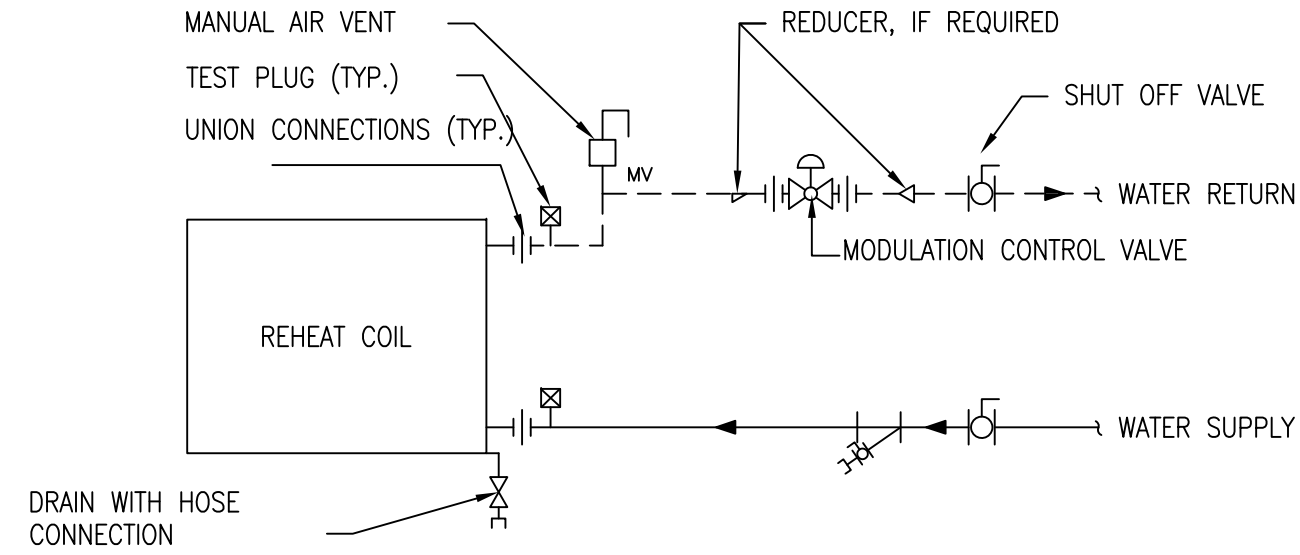
NTS



- DESIGNER'S NOTE:
1. COORDINATE WITH HVAC DESIGN MANUAL.
 2. CONSIDER REMOVING MANUAL BYPASS, BECAUSE CONTROL VALVES ARE REASONABLY RELIABLE AND THE PRESENCE OF THE BYPASS MAY ENCOURAGE POSTPONING REPAIRS, THEREBY LESSENING THE EFFECTIVENESS OF VARIABLE SPEED PUMPING SYSTEMS

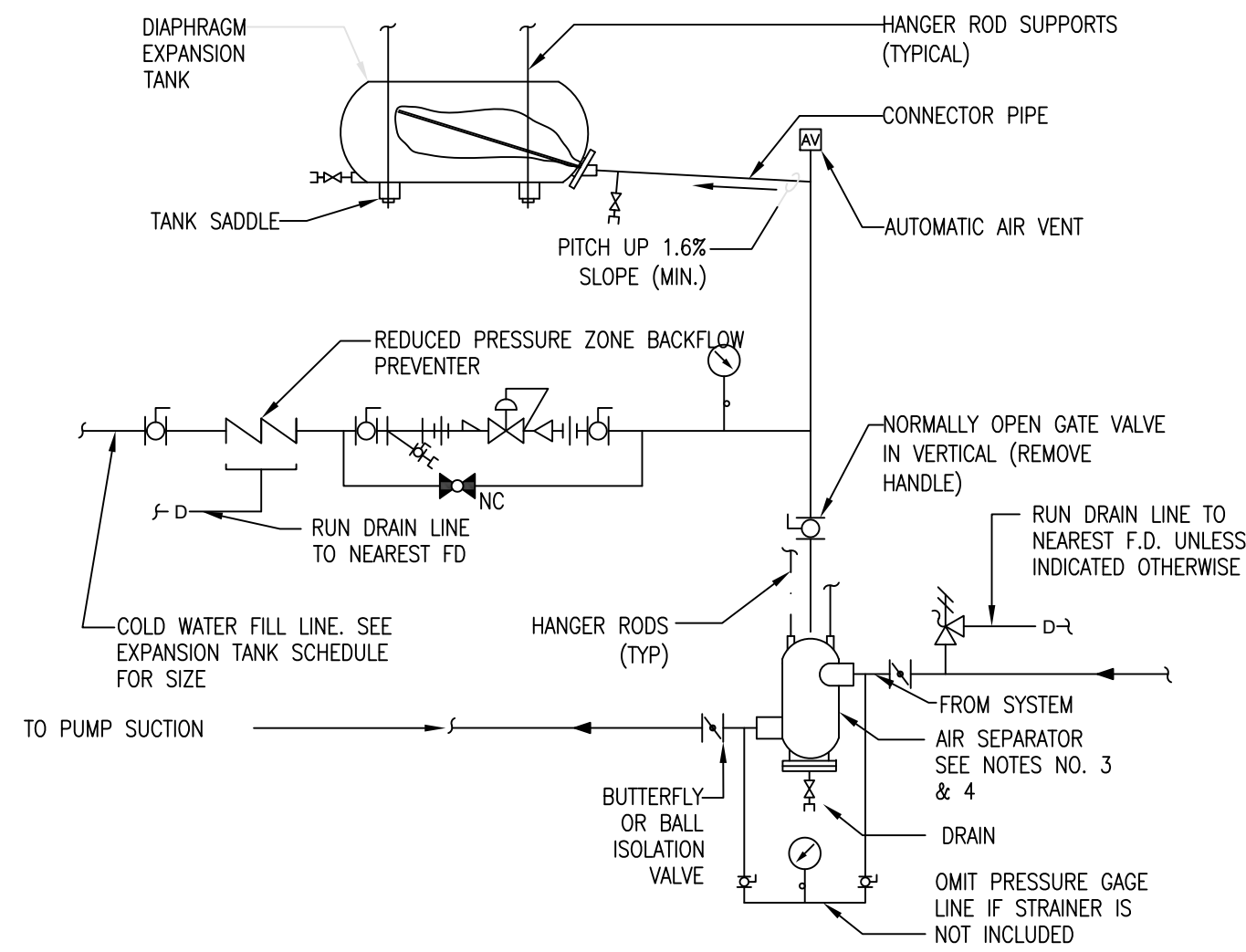
2 PREHEAT COIL (HOT WATER) - PIPING CONNECTIONS

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3 REHEAT COIL PIPING CONNECTIONS

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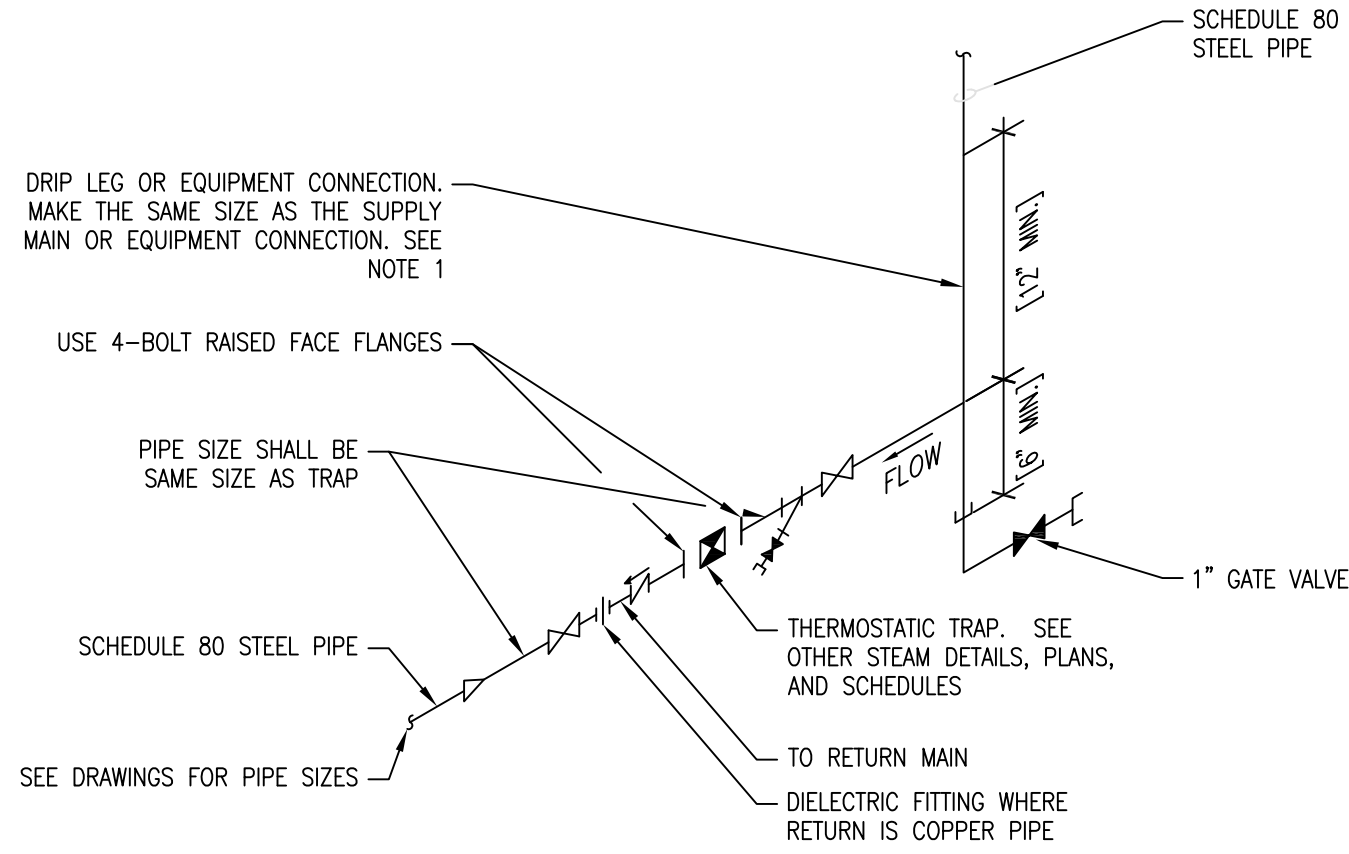


- NOTES:
1. SEE EXPANSION TANK SYSTEM SCHEDULE FOR COMPONENT SIZES.
 2. RELIEF VALVE FOR CHILLED WATER SYSTEM IS SHOWN. OMIT WHEN RELIEF VALVE IS SHOWN ON HEAT EXCHANGER DETAIL & SYSTEM IS USED ONLY FOR HOT WATER HEATING.
 3. PROVIDE STRAINER IN AIR SEPARATOR WHEN INDICATED IN EXPANSION TANK SCHEDULE.
 4. FOR HOT WATER SYSTEMS 50mm [2"] AND SMALLER AND CHILLED WATER SYSTEMS USE IN-LINE AIR PURGER IN LIEU OF AIR SEPARATOR.
 5. SET PRESSURE REDUCING VALVE SO PRESSURE AT HIGHEST POINT IN SYSTEM HAS A MINIMUM OF 28kPa [4 PSIG]

DESIGNER'S NOTE:
VALVES SHALL BE INDICATED ON EITHER SIDE OF AIR SEPARATOR AS REQUIRED BY CLOSURE OF VALVES SERVING ADJACENT EQUIPMENT. WHERE CHARGING OF TANK IS PROPOSED PROVIDE NECESSARY TAPPINGS. PROVIDE AND SHOW A LOW WATER ALARM ON CHARGED SYSTEMS TO INDICATE NO WATER IN TANK.

4 HORIZONTAL EXPANSION TANK - PIPING CONNECTIONS

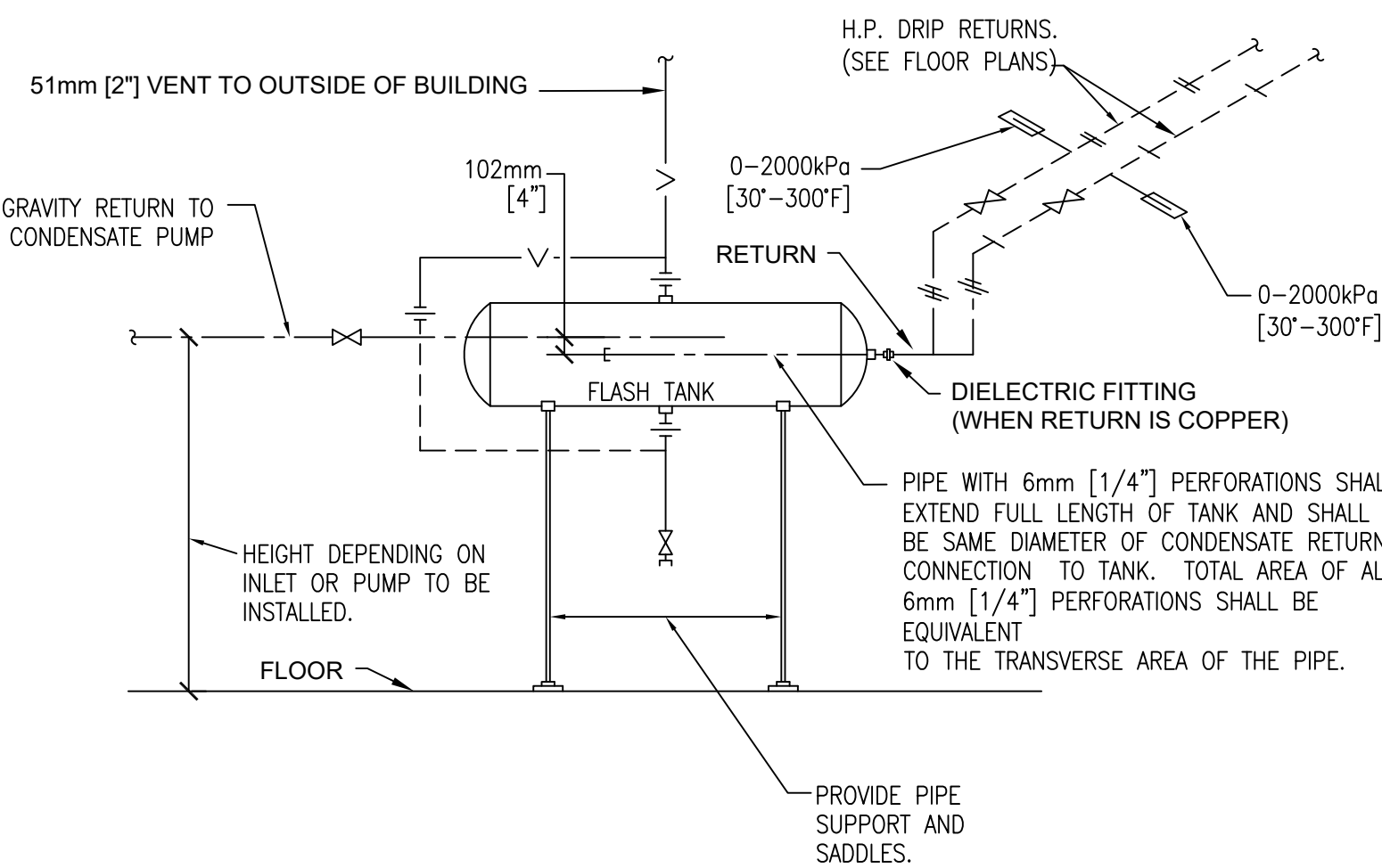
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- NOTE:
1. ALL DRIP POINTS ON STEAM MAINS SHALL BE PROVIDED WITH A 300mm [12"] MINIMUM HIGH DRIP LEG FROM BOTTOM OF STEAM MAIN TO TRAP INLET. DRIP LEG SHALL HAVE 150mm [6"] SCALE POCKET BELOW TRAP INLET.

5 THERMOSTATIC STEAM TRAP ASSEMBLY

NTS

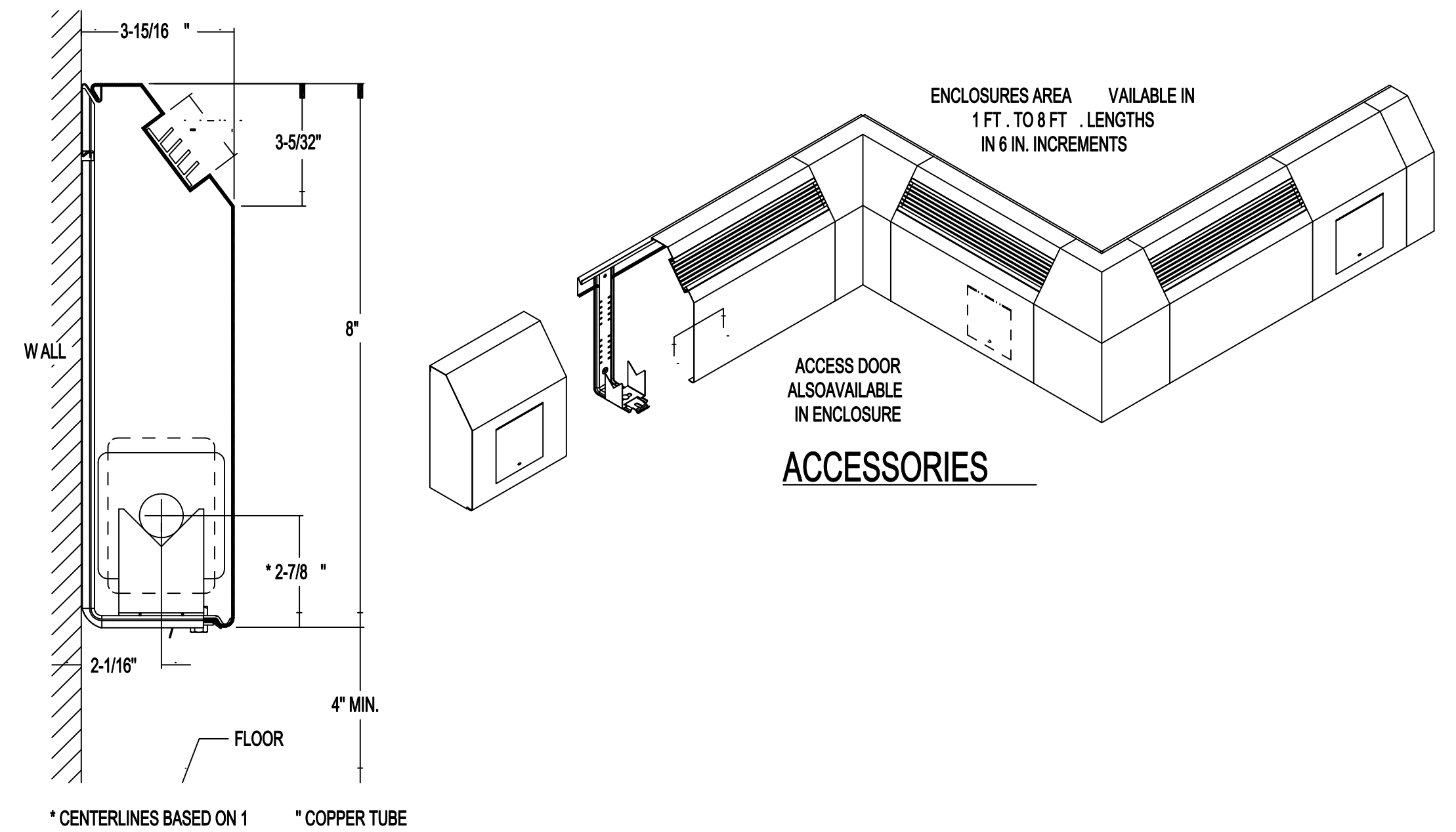


FLASH TANK SCHEDULE		
CONDENSATE PUMP CAPACITY- [GPM]	APPROX. CAPACITY OF FLASH TANK-L [GALLONS]	SIZE OF FLASH TANK- mm [in]
0 THRU 15	61[16]	356mm DIA X 610mm LONG [14 DIA. X 24 LONG]
16 THRU 22	91[24]	365mm DIA X 914mm LONG [14 DIA. X 36 LONG]
23 THRU 30	117[31]	406mm DIA X 914mm LONG [16 DIA. X 36 LONG]
31 THRU 37	140[37]	406mm DIA X 1067mm LONG [16 DIA. X 42 LONG]

- DESIGNER'S NOTE:
1. INDICATE THE HEIGHT ON FLOOR PLANS AND/OR SECTIONS. PROVIDE A FLASH TANK FOR EACH CONDENSATE PUMP, THAT SERVES HPR CONDENSATE.
 2. FOR FLASH STEAM RECOVERY, FLASH TANK TO BE VERTICAL TYPE. PROVIDE BACK PRESSURE VALVE AND SAFETY RELIEF VALVE AT FLASH STEAM LINE. PROVIDE TRAP AT BOTTOM CONDENSATE DISCHARGE LINE. NO VENT INTERCONNECTION BETWEEN CONDENSATE DISCHARGE AND THE FLASH STEAM LINE.

6 TYPICAL CONNECTIONS TO FLASH TANK

NTS



7 FIN TUBE RADIATOR INSTALLATION

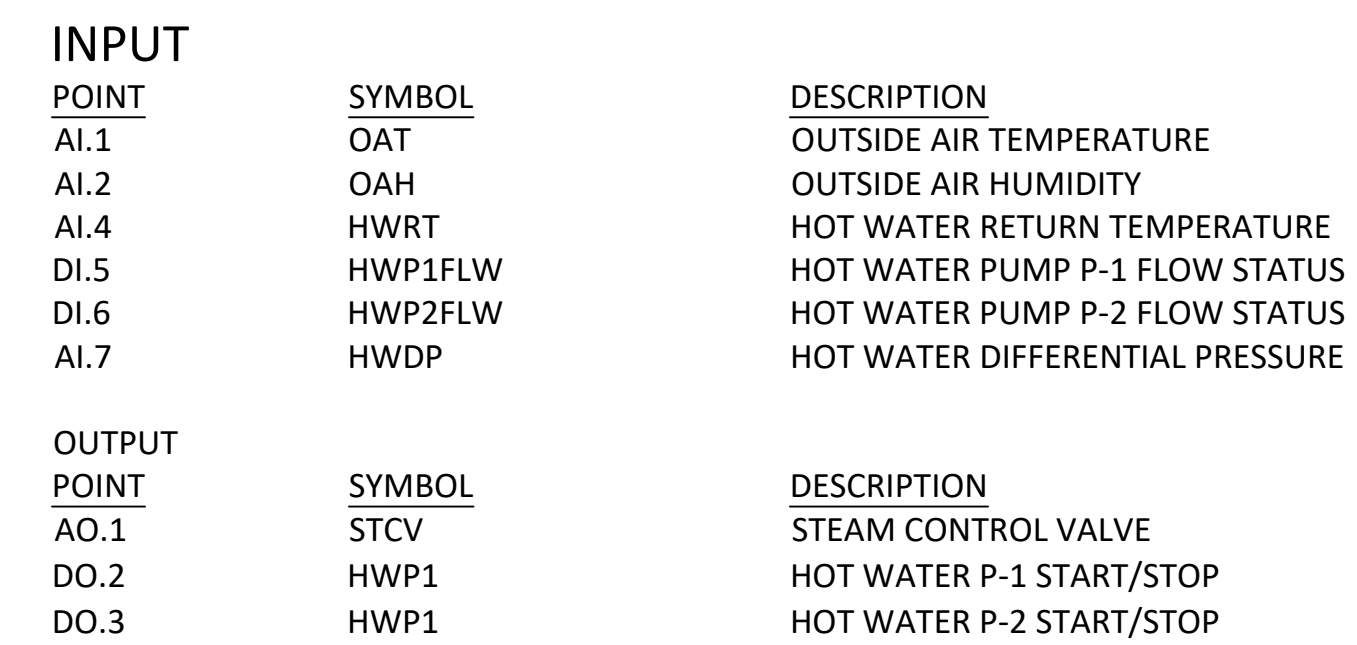
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Revisions:	Date	Revisions Continued:	Date	VA STATION VA HEALTH CARE Defining EXCELLENCE in the 21st Century Jesse Brown VA Medical Center 820 S. Damen Ave. Chicago, IL, 60612	TO NGA T. NGUYEN 062-050406 LICENSED PROFESSIONAL ENGINEER OF THE STATE OF ILL. NO. 09-0000000 EXPIRES NOV. 30, 2025	Office of Construction and Facilities Management VA U.S. Department of Veterans Affairs	Drawing Title MECHANICAL DETAILS I	Phase BID DOCUMENTS	Project Title AHU-AC21 REPLACEMENT SERVING NUTRITION FOOD AND SERVICE	Project Number 537-26-200	Office of Planning and Construction Department of Veterans Affairs
Approved Project Director		FULLY SPRINKLERED		JESSE BROWN VAMC	Location	Date: October 30, 2025	Checked: D. Moreau	Prepared: A. Nguyen	Building Number 1-B	Drawing Number M-4	Dwg. 9 of 16

FULLY SPRINKLERED





3 TO 5 PIPE DIAMETERS

SPRING HANGER

DISCHARGE

MOUNT PIPE HANGER AT EACH PIPE DROP TO AVOID STRAIN ON PUMP (TYPICAL) SEE SPECIFICATIONS

TRIPLE DUTY VALVE

PIPE INCREASER

BASE MOUNTED PUMP

ANCHOR PUMP

PER MANUF. RECOMMENDATION

6" THICK CONCRETE BASE

ALTERNATE LOCATION FOR TRIPLE DUTY VALVE

SPRING HANGER

LOCATE ALL PIPE SUPPORTS SUPPORT TYPE IN ACCORD WITH SPECIFICATIONS

SUCTION

1/4" COPPER TUBES

PRESSURE GAUGE

HYDRONIC INDICATOR W/MULTI PORT GAUGE

BUTTERFLY VALVE (BFV)

FLEXIBLE CONNECTION DOUBLE SPHERICAL W/ CONTROL RODS (TYPICAL)

SUCTION DIFFUSER WITH STRAINER

HOSE END DRAIN VALVE

ADJUSTABLE PIPE SUPPORT FOOT

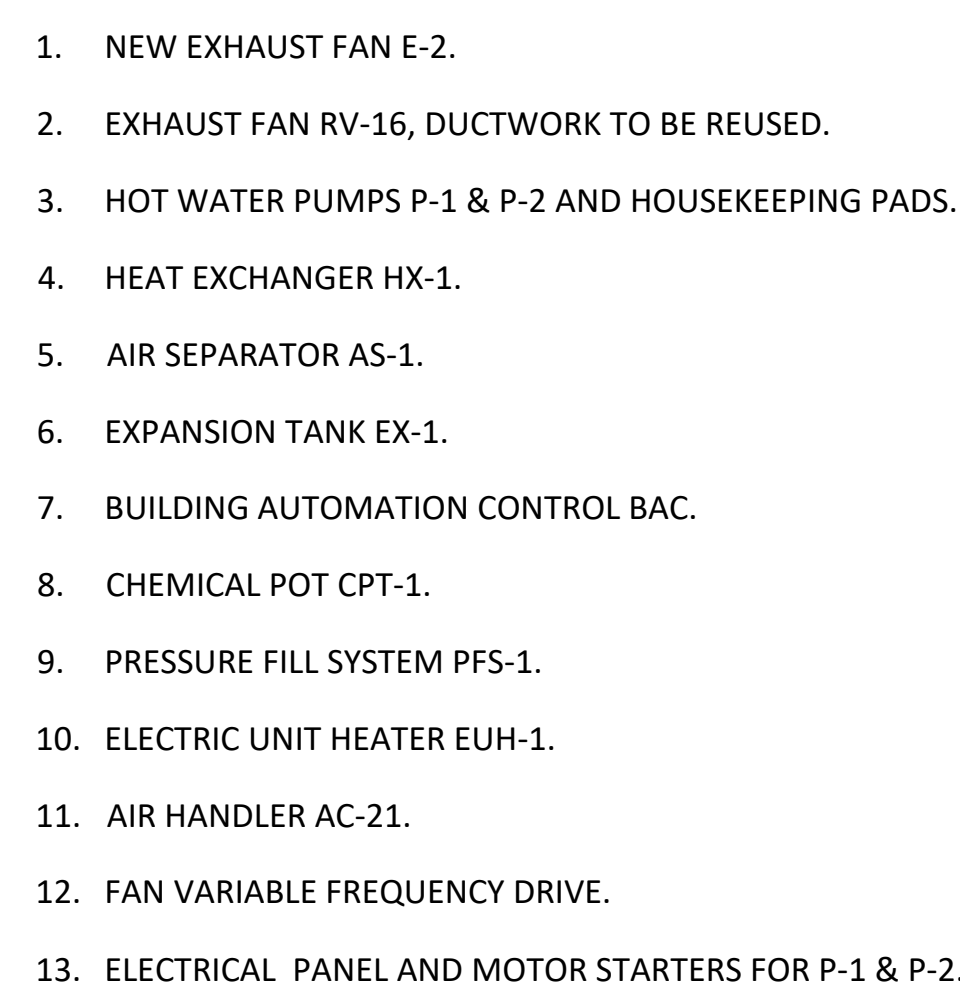
INERTIA BASE WITH VIBRATOR ISOLATORS. SEE SPECIFICATIONS

The diagram illustrates a hot water system with the following components and connections:

- Hot Water Pumps:** Two pumps, (P-1) and (P-2), are located at the bottom. Each is mounted on a 6" concrete pad and includes a flex line and a triple duty valve. Pump (P-1) is labeled "HOT WATER PUMP" and pump (P-2) is labeled "HOT WATER PUMP".
- Heat Exchanger:** A "NEW HEAT EXCHANGER (HX-1)" is located on the left. It has a "CONVERTOR" on top and a "PRESSURE RELIEF" valve on the right. It is connected to a "4" HWS" line and a "4" HWR" line.
- Expansion Tank:** An "EXPANSION TANK EX-1" is located at the top right. It is connected to a "1" EXPANSION LINE" and a "2 1/2" line".
- Air Separator:** An "AIR SEPARATOR AS-1" is located on the right. It is connected to a "4" HWS" line and a "2 1/2" line".
- Condensate Return:** An "EXISTING VACUUM STEAM CONDENSATE RETURN" is located on the far left. It is connected to a "2 1/2" HWS" line.
- Piping and Valves:** The system includes various piping sizes (5", 6", 4", 2 1/2", 1") and valves (gate valves, check valves, pressure relief valve, triple duty valve). A "MANUAL BYPASS" is shown near the expansion tank.
- Instrumentation:** The diagram includes symbols for pressure gauges, flow indicators, and level indicators.

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FULLY SPRINKLERED



1. CONTRACTOR SHALL REPLACE THE EXISTING AIR HANDLER AC-1 WITH A PERFORMANCE AS REQUIRED IN THE AHU AC-21 SCHEDULE. TO DWG M-2. INFORMATION ON THE REQUIRED PERFORMANCE FOR E-2 AND RV-16 IS ALSO FOUND ON DWG M-2. THE CONTRACTOR IS REQUIRED TO REVIEW AND COMPLY WITH SPEC SECTIONS:

23 07 11 HVAC AND BOILER PLANT INSULATION
23 21 13 HYDRONIC PIPING
23 21 23 HYDRONIC PUMPS
23 22 13 STEAM AND CONDENSATE HEATING PIPING
23 74 13 PACKAGED, OUTDOOR, CENTRAL-STATION AIR-HANDLING
UNITS

2. PREPARE A CONSTRUCTION SCHEDULE AND SUBMIT PROPOSED SCHEDULE TO THE COR FOR REVIEW AND APPROVAL. NO WORK CAN BE STARTED UNTIL THE SCHEDULE IS APPROVED BY THE COR.

3. DO NOT INTERRUPT EXISTING UTILITIES SERVICING OCCUPIED OR USED FACILITIES, EXCEPT WHEN APPROVED IN WRITING BY THE COR AND THEN ONLY AFTER TEMPORARY UTILITY SERVICES HAVE BEEN APPROVED AND PROVIDED.

[illegible]

Drawing Title
GENERAL LAYOUT FOR NEW AHU AC-21
Approved Project Director

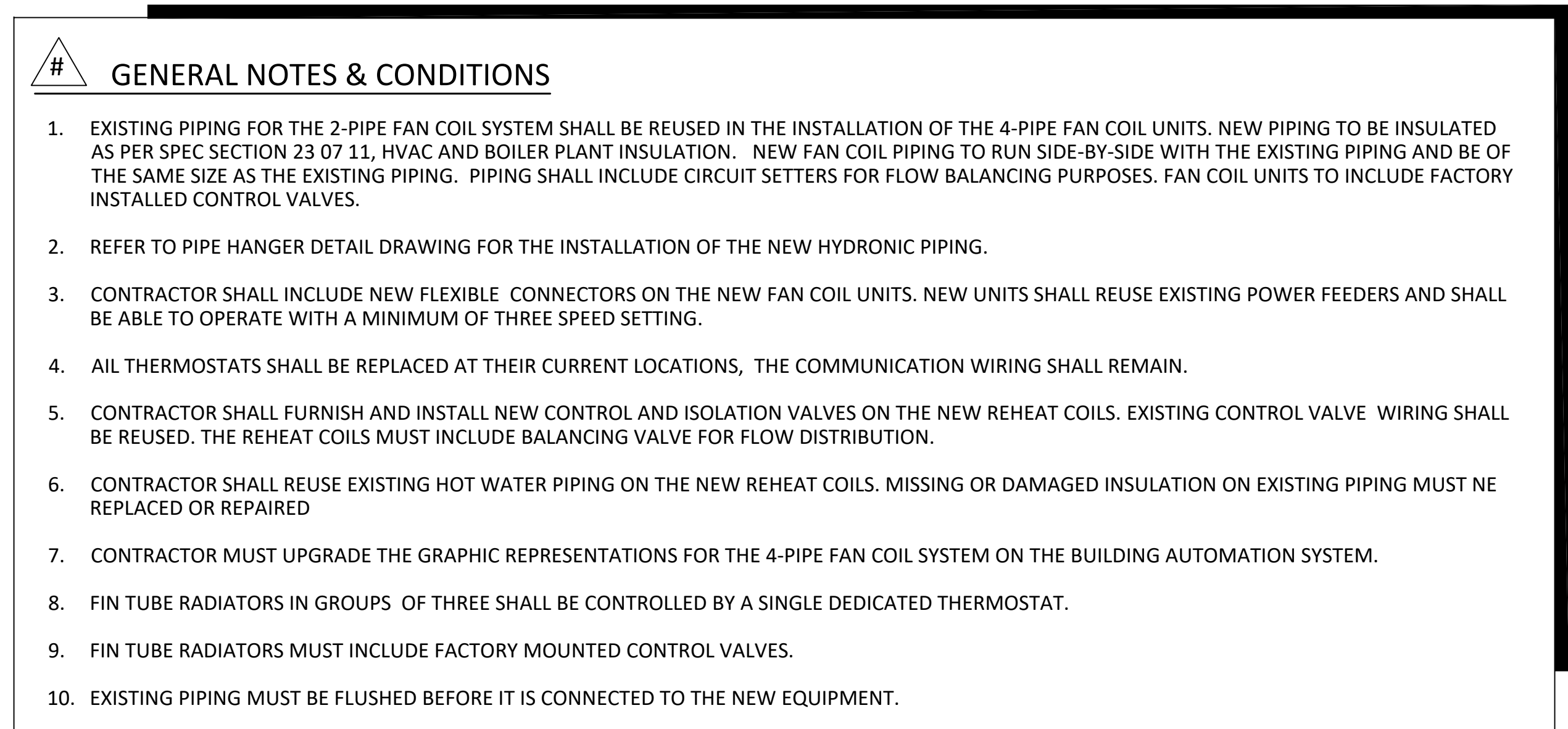
Phase	BID DOCUMENTS
	FULLY SPRINKLERED

Project Title		
AHU-AC21 REPLACEMENT SERVING NUTRITION FOOD AND SERVICE		
Location		
JESSE BROWN VAMC		
Date:	Checked:	Prepared:
October 30, 2025	D. Moreau	A. Nguyen

Project Number 537-26-200 Building Number 1-B Drawing Number M-7	Dwg. 12 of 16

Office of
Planning and
Construction

 Department of
Veterans Affairs



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[illegible]

TEMPERATURE CONTROL GENERAL NOTES

4. WALL THERMOSTATS AND/OR TEMPERATURE SENSORS SHALL BE INSTALLED AT 48" TO 54" ABOVE FLOOR ADJACENT TO OR ABOVE LIGHT CONTROL DEVICE

5. UNLESS OTHERWISE NOTED, TEMPERATURE CONTROL WIRING/PIPING SHALL BE CONCEALED ABOVE CEILINGS, IN WALLS, OR INSIDE CHASES.
2. PROVIDE FIRE STOPPING, WITH NO EXCEPTION, WHERE FIRE AND/OR SMOKE WALLS ARE PENETRATED.

6. CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING ADEQUATE SUPPORTING SYSTEMS FOR TEMPERATURE CONTROL W IRING/PIPING, EQUIPMENT, AND ACCESSORIES.

7. SEE SPECIFICATION FOR CONTROLS AND INSTRUMENTATION REQUIREMENTS AND THE SEQUENCE OF OPERATION FOR EQUIPMENT

8. TEMPERATURE CONTROL CABLING SHALL BE LOW CAPACITANCE 24 GA TWISTED SHIELDED PAIR WITH HARD JACKET

9. ALL CONTROLS CABLING SHALL BE PLENUM RATED

TEMPERATURE CONTROL LEGEND

SYMBOL	DESCRIPTION
AFV	ABOVE FINISHED FLOOR
RTU#	ROOFTOP UNIT
BLN	BUILDING LEVEL NETWORK
MAU#	MAKEUP AIR UNIT
MBC	MAIN BUILDING CONTROLLER
NOC	NORMALLY CLOSED
NOP	NORMALLY OPEN
TYP.	TYPICAL
UH#	UNIT HEATER
VFD	VARIABLE FREQUENCY DRIVE
VSD	VARIABLE SPEED DRIVE
T	THERMOSTAT
TS	TEMPERATURE SENSOR
CO	CO SENSOR
NOC2	N2 O2 SENSOR
Tr	REVERSE ACTING THERMOSTAT

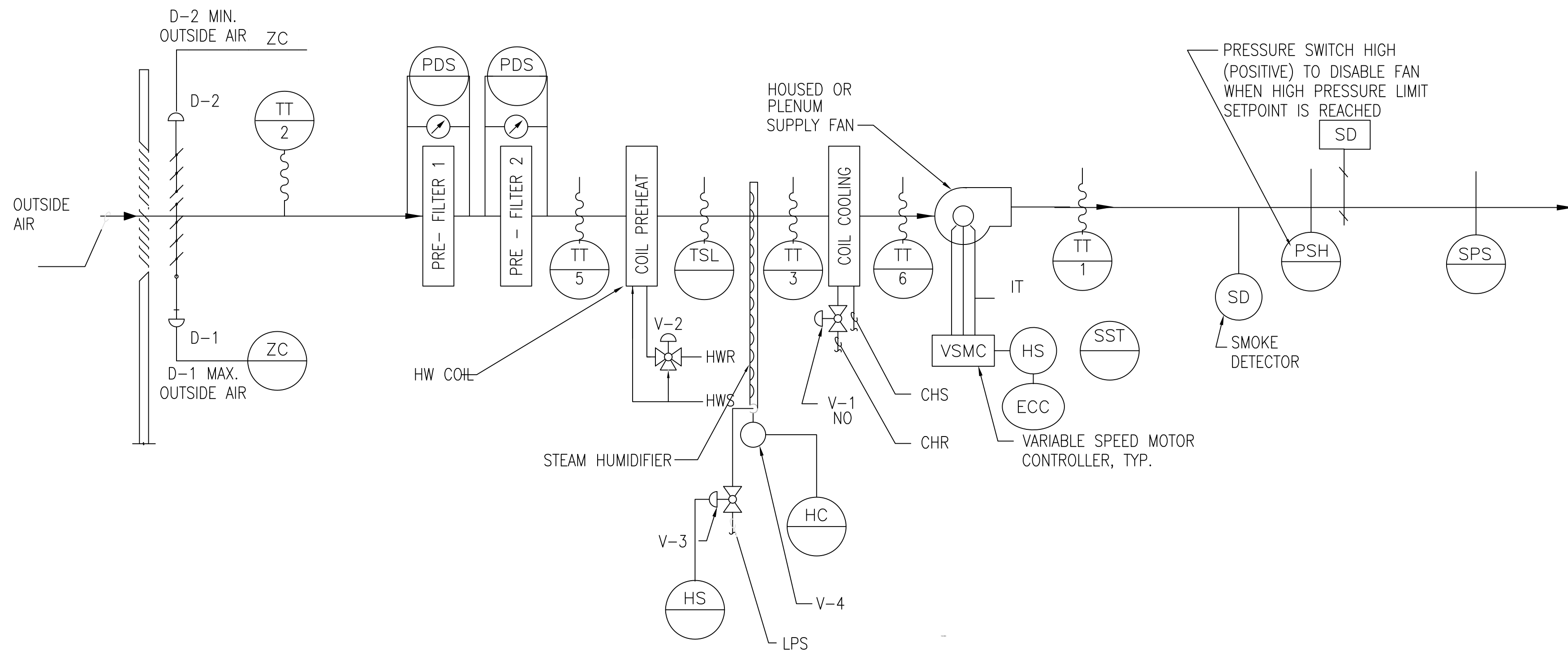
CONTROL SYMBOLS

	ROOM THERMOSTAT/TRANSMITTER – WALL MOUNT		TEMPERATURE SWITCH, HIGH (FREEZESTAT)
	ROOM HUMIDISTAT (MOISTURE)/TRANSMITTER – WALL MOUNT		LEVEL CONTROLLER
	TEMPERATURE TRANSMITTER		LEVEL TRANSMITTER
	TEMPERATURE TRANSMITTER, AVERAGING ELEMENT		PRESSURE SWITCH HIGH
	MOISTURE (HUMIDITY) TRANSMITTER		PRESSURE SWITCH LOW
	PRESSURE TRANSMITTER		ELECTRONIC TO PNEUMATIC TRANSDUCER
	STATIC PRESSURE SENSOR		OCCUPANCY SENSOR
	FLOW TRANSMITTER		LOCAL TEMPERATURE CONTROL PANEL
	CURRENT TRANSMITTER		HVAC CONTROL PANEL
	PRESSURE DIFFERENTIAL TRANSMITTER		VARIABLE SPEED MOTOR CONTROLLER
	PRESSURE DIFFERENTIAL SWITCH		TEMPERATURE SENSING ELEMENT FOR TRANSMITTING TEMPERATURE TO EMCS (PROVIDE 12 INCHES [200mm] MINIMUM LENGTH IN DUCT WHEN SPACE PERMITS.)
	HAND SWITCH (HAND-OFF-AUTO SWITCH)		SENSOR WITH AVERAGING ELEMENT TO TRANSMIT TEMPERATURE TO EMCS
	VALVE OR DAMPER POSITION CONTROLLER		MOTOR STARTER
	TEMPERATURE SWITCH, LOW (FREEZESTAT)		ELECTRIC OPERATED CONTROL DAMPER/OR VALVE

2. WHEN ADDITIONAL TEMPERATURE CONTROL PANELS REQUIRING 120V ELECTRICAL POWER ARE PROVIDED, COORDINATE LOCATION WITH THE ELECTRICAL CONTRACTOR.

SYSTEM ARCHITECTURE NOTES

1. THE INFORMATION CONTAINED WITHIN THIS DIAGRAM IS FOR GENERAL ARRANGEMENT OF THE DIRECT DIGITAL CONTROL SYSTEM ONLY, THE ACTUAL SYSTEM ARCHITECTURE SHALL BE DESIGNED FULLY BY THE TEMPERATURE CONTROLS CONTRACTOR IN CONSULTATION WITH THE VENDOR



100% OUTSIDE AIR HANDLING UNIT AC-21 FOR CAFETERIA

NTS

SEQUENCE OF OPERATION FOR VARIABLE AIR VOLUME AIR HANDLING UNIT WITH MINIMUM OUTSIDE AIR

1. GENERAL

UNIT IS NORMALLY STARTED AND STOPPED REMOTELY AT THE ECC. H-0-A SWITCH SHALL BE KEPT IN THE "AUTO" POSITIO2. "HAND" AND "OFF" POSITIONS SHALL BE USED ONLY FOR MAINTENANCE. WHEN THE UNIT IS "OFF" D-1, D-3, SHALL BE FULLY CLOSED. WHEN THE UNIT IS "ON" D-1, SD-1 AND SD-2 SHALL BE FULLY OPEN. D-2 AND D-3 SHALL MODULATE IN ACCORDANCE WITH THE FOLLOWING SEQUENCE:

2. TEMPERATURE CONTROL

2.1 SUPPLY AIR TEMPERATURE, SENSED BY TT-1, SHALL BE MAINTAINED AT SETPOINT VIA DIGITAL CONTROL PANEL BY MODULATING V-1 OR D-1 AND D-2 AND V-2 IN SEQUENCE.

2.2 WHEN THE TEMPERATURE OF THE OUTSIDE AIR, SENSED BY TT-2, IS ABOVE 75°F (ADJ), THE DIGITAL CONTROL PANEL SHALL PREVENT THE MODULATION OF D-1 AND D-2 AND SHALL ASSUME THE MINIMUM OUTSIDE AIR POSITION (D-2 FULLY OPENED AND D-1 FULLY CLOSED). THE DIGITAL CONTROL PANEL SHALL MODULATE V-1 TO MAINTAIN THE SUPPLY AIR TEMPERATURE, SENSED BY TT-1.

2.3 WHEN THE TEMPERATURE OF THE OUTSIDE AIR, SENSED BY TT-2, IS BETWEEN 65°F (18.3°C) AND THE SUPPLY AIR TEMPERATURE SENSED BY TT-1, D1 AND D2 SHALL BE FULLY OPEN (MAXIMUM OUTSIDE AIR POSITION). THE DIGITAL CONTROL PANEL SHALL MODULATE V-1 TO MAINTAIN THE SUPPLY AIR TEMPERATURE, SENSED BY TT-1.

2.4 WHEN THE TEMPERATURE OF THE OUTSIDE AIR, SENSED BY TT-2, IS BELOW THE SUPPLY AIR TEMPERATURE, SENSED BY TT-1, DAMPERS D-1, AND D-2 SHALL MODULATE TO MAINTAIN THE SCHEDULED SUPPLY AIR TEMPERATURE. IF D-1 IS OPEN AND D-2 IS CLOSED TO MINIMUM OUTSIDE AIR, V-2 SHALL MODULATE OPEN TO MAINTAIN THE SUPPLY AIR TEMPERATURE, SENSED BY TT-1.

3. AIR FLOW CONTROL

3.1 THE SUPPLY AIR FLOW SHALL BE CONTROLLED BY THE DIGITAL CONTROL PANEL MODULATING THE SUPPLY FAN VARIABLE SPEED MOTOR CONTROL TO MAINTAIN 1.0" [25mm] OF DUCT STATIC PRESSURE (FIELD ADJUSTABLE), SENSED BY SPS-1. RESET STATIC PRESSURE BASED ON ACTUAL BUILDING LOAD BY POLLING ALL ATU

3.2 THE DIGITAL CONTROL PANEL, USING TOTAL SUPPLY AIR AND EXHAUST AIR FROM E-2 AIR FLOW SIGNALS, SHALL RESET THE EXHAUST AIR FAN TO MAINTAIN A CONSTANT AIR FLOW DIFFERENCE BETWEEN THE SUPPLY AIR AND THE EXHAUST AIR EQUAL TO MINIMUM OUTSIDE AIR.

3.3 USING HIGH PRESSURE SENSOR SPS-2 LOCATED AT THE SUPPLY FAN DISCHARGE, SHALL PREVENT THE SUPPLY FAN FROM DEVELOPING OVER 3" STATIC PRESSURE (FIELD ADJUSTABLE). IF STATIC PRESSURE AT SPS-2 DOES EXCEED 3" THE SUPPLY AIR FAN SHALL STOP. SPS-2 SHALL BE HARDWIRED TO THE SUPPLY FAN VSMC AND UNIT SHALL BE SHUTDOWN IN HAND/AUTO OR BYPASS MODE. SPS-2 WILL REQUIRE MANUAL RESET OF THE DEVICE.

4. HUMIDITY CONTROL

4.1 WHEN THE DIGITAL CONTROL PANEL IS NOT CALLING FOR HUMIDITY, SENSED BY RETURN AIR HUMIDITY H-1, 2-WAY "ON-OFF" CONTROL VALVE V-3 SHALL REMAIN CLOSED. WHEN THE DIGITAL CONTROL PANEL IS CALLING FOR HUMIDITY, V-3 SHALL REMAIN OPEN.

4.2 RETURN AIR HUMIDITY SHALL BE MAINTAINED AT SETPOINT OF 35% RH (ADU) VIA DIGITAL CONTROL PANEL BY MODULATING CONTROL VALVE V-4 TO MAINTAIN THE DESIRED HUMIDITY. THE DCP SHALL OVERRIDE THIS CONTROL TO MAINTAIN HUMIDITY OF 80% AS SENSED BY H-2. DCP SHALL CLOSE VALVE V-3 WHENEVER THE SUPPLY FAN IS OFF. VALVE V-4 SHALL BE INTERLOCKED WITH A TEMPERATURE SWITCH TO KEEP THE HUMIDIFIER OFF UNTIL CONDENSATE TEMPERATURE APPROACHES STEAM TEMPERATURE.

5. FREEZE PROTECTION

5.1 IF THE AIR TEMPERATURE AS SENSED BY TT-3 FALLS BELOW 45°F [7°C], AN ALARM SIGNAL SHALL INDICATE AT THE DCP AND ECC. IF THIS TEMPERATURE FALLS BELOW 40°F [4.4°C], AS SENSED BY THE TSL THE SUPPLY AND RETURN FANS SHALL SHUT DOWN AND A CRITICAL ALARM SHALL INDICATE AT THE DIGITAL CONTROL PANEL AND ECC. TSL SHALL BE HARDWIRED TO THE SUPPLY FAN FUD AND UNIT SHALL BE SHUTDOWN IN HAND/AUTO OR BYPASS MODE. TSL WILL REQUIRE MANUAL RESET AT THE DEVICE.

6. AUTOMATIC SHUTDOWN/RESTART

6.1 WHEN SMOKE IS DETECTED BY DUCT SMOKE DETECTOR, SD, THE SUPPLY AND RETURN FANS SHALL SHUT "OFF" AND AN ALARM SIGNAL SHALL BE TRANSMITTED TO THE FIRE ALARM SYSTEM. ALL SMOKE DAMPERS IN THE SUPPLY AND RETURN DUCTS SHALL CLOSE.

6.2 EXHAUST FANS SERVING AREA OF THE SUPPLY FAN SHALL CONTINUE TO RUN. SUPPLY AND RETURN FANS SHALL RESTART AND SMOKE DAMPERS SHALL OPEN WHEN FIRE ALARM CIRCUIT IS RESET.

7. EMERGENCY CONSTANT SPEED OPERATION

7.1 UPON FAILURE OF THE VSMC, THE SUPPLY AND RETURN FANS SHALL BE STARTED/STOPPED MANUALLY AT THE DIGITAL CONTROL PANEL OR THE ECC THROUGH THE BY-PASS STARTER. FANS SHALL THEN BE OPERATED AT CONSTANT SPEED.

[illegible]

POINTS LIST FOR AIR HANDLING UNIT WITH 100% OUTSIDE AIR

ITS

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[illegible]

FAN COIL SEQUENCE OF OPERATION

SEQUENCE OF OPERATION
MODE SELECTION: THE BUILDING'S CENTRAL PLANT DETERMINES WHETHER THE TWO-PIPE SYSTEM WILL SUPPLY HOT OR COLD WATER.
TEMPERATURE SENSING: A SENSOR IN THE ROOM MEASURES THE ACTUAL SPACE TEMPERATURE AND SENDS THIS INFORMATION TO THE FCU'S CONTROLLER.
FAN CONTROL: BASED ON THE DIFFERENCE BETWEEN THE ROOM'S TEMPERATURE AND THE SETPOINT, THE CONTROLLER WILL TURN THE FAN ON AND ADJUST ITS SPEED.
IF THE TEMPERATURE IS SIGNIFICANTLY DIFFERENT FROM THE SETPOINT, THE FAN RUNS AT HIGH SPEED TO QUICKLY ADJUST THE AIR TEMPERATURE.
AS THE ROOM TEMPERATURE GETS CLOSER TO THE SETPOINT, THE FAN SPEED DECREASES.

FAN COIL SHALL OPERATE ON A SCHEDULE AS SET BY BAS. FAN SHALL RUN CONTINUOUSLY IN OCCUPIED MODE. FAN STATUS SHALL BE MONITORED AND AN ALARM MESSAGE SHALL BE GENERATED IN THE EVENT THE UNIT FAILS TO RUN BETWEEN THE RANGE OF 70°-75° SPACE TEMPERATURE BOTH V-1 & V-2 SHALL BE CLOSED. UPON RISE IN TEMPERATURE ABOVE 75° V-2 SHALL MODULATE OPEN TO MAINTAIN 75° F. UPON FALL IN TEMPERATURE BELOW 70° F. HEATING VALVE V-1 SHALL MODULATE TO OPEN TO MAINTAIN 70° F. UNITS SHALL INCLUDE A MINIMUM OF THREE SPEEDS WITH SPEED SELECTION BY THE BAS AT THE ECC.

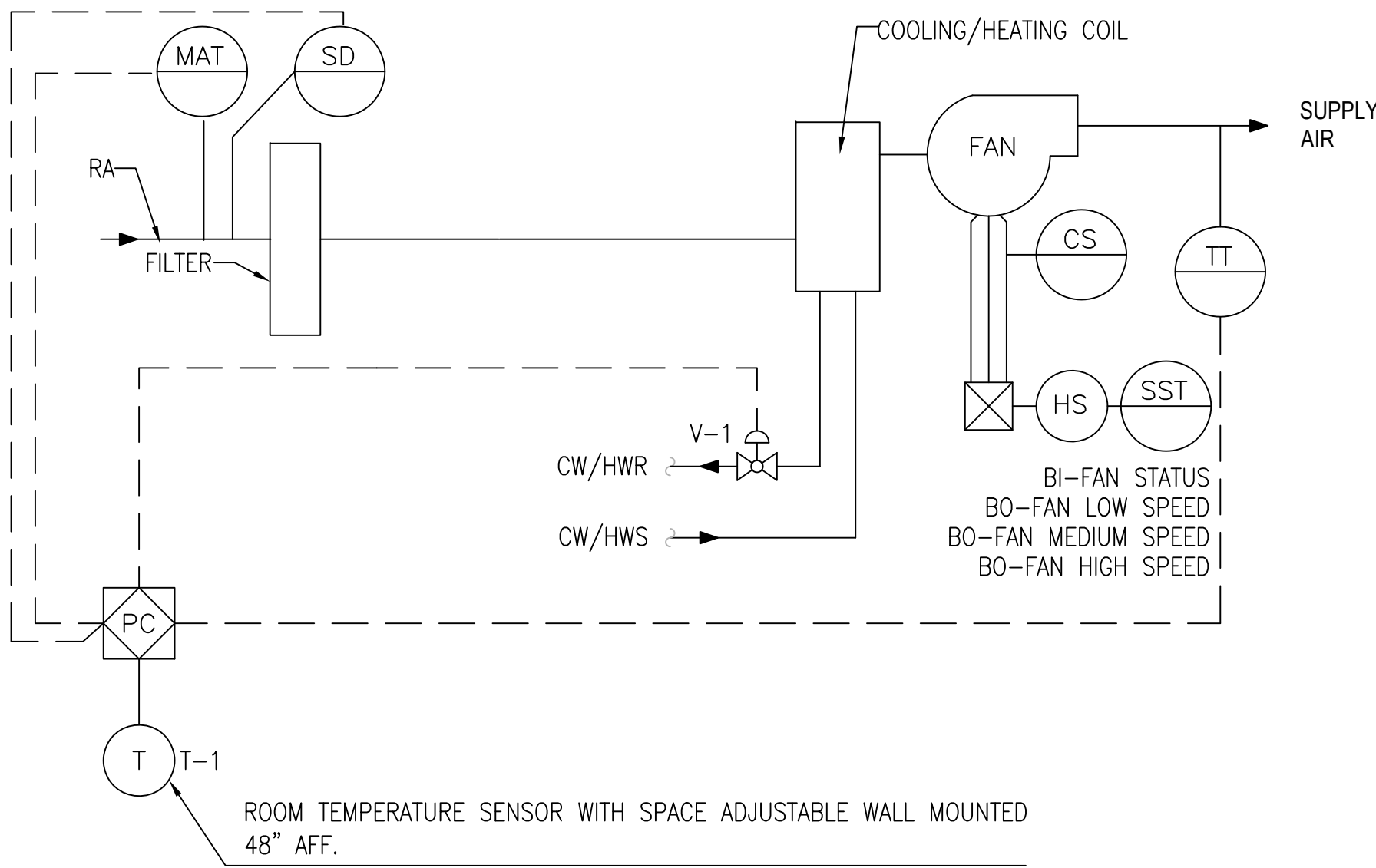
RUN CONDITIONS

THE UNITS SHALL RUN ACCORDING TO A USER DEFINABLE TIME SCHEDULE IN THE FOLLOWING MODES:

- OCCUPIED MODE: THE UNITS SHALL CONTINUOUSLY RUN TO MAINTAIN
A 74° F (ADJ.) COOLING SETPOINT
A 70° F (ADJ.) HEATING SETPOINT
A 60% RH (ADJ.) HUMIDITY SETPOINT
UNOCCUPIED MODE NIGHT SETBACK: THE UNITS SHALL CYCLE TO
A 80° F (ADJ.) COOLING SETPOINT
A 62° F (ADJ.) HEATING SETPOINT
A 60% RH (ADJ.) HUMIDITY SETPOINT

ALARMS SHALL BE PROVIDED AS FOLLOWS:
HIGH SPACE TEMPERATURE: IF THE SPACE TEMPERATURE IS GREATER THAT THE COOLING SETPOINT BY A USER DEFINABLE AMOUNT (ADJ.) LOW SPACE TEMP: IF THE SPACE TEMPERATURE IS LESS THAN THE HEATING SETPOINT BY A USER DEFINABLE AMOUNT (ADJ.)

FREEZE PROTECTION:
THE UNIT SHALL SHUTDOWN AND GENERATE AN ALARM UPON RECEIVING A FREEZESTAT STATUS. WHENEVER THE FREEZESTAT IS ON, THE WATER COIL VALVE SHALL MODULATE OPEN TO MAINTAIN THE MIXED AIR TEMPERATURE AT 70° F (ADJ.) THE FAN COIL FRESH AIR DAMPER SHALL FULLY CLOSE,

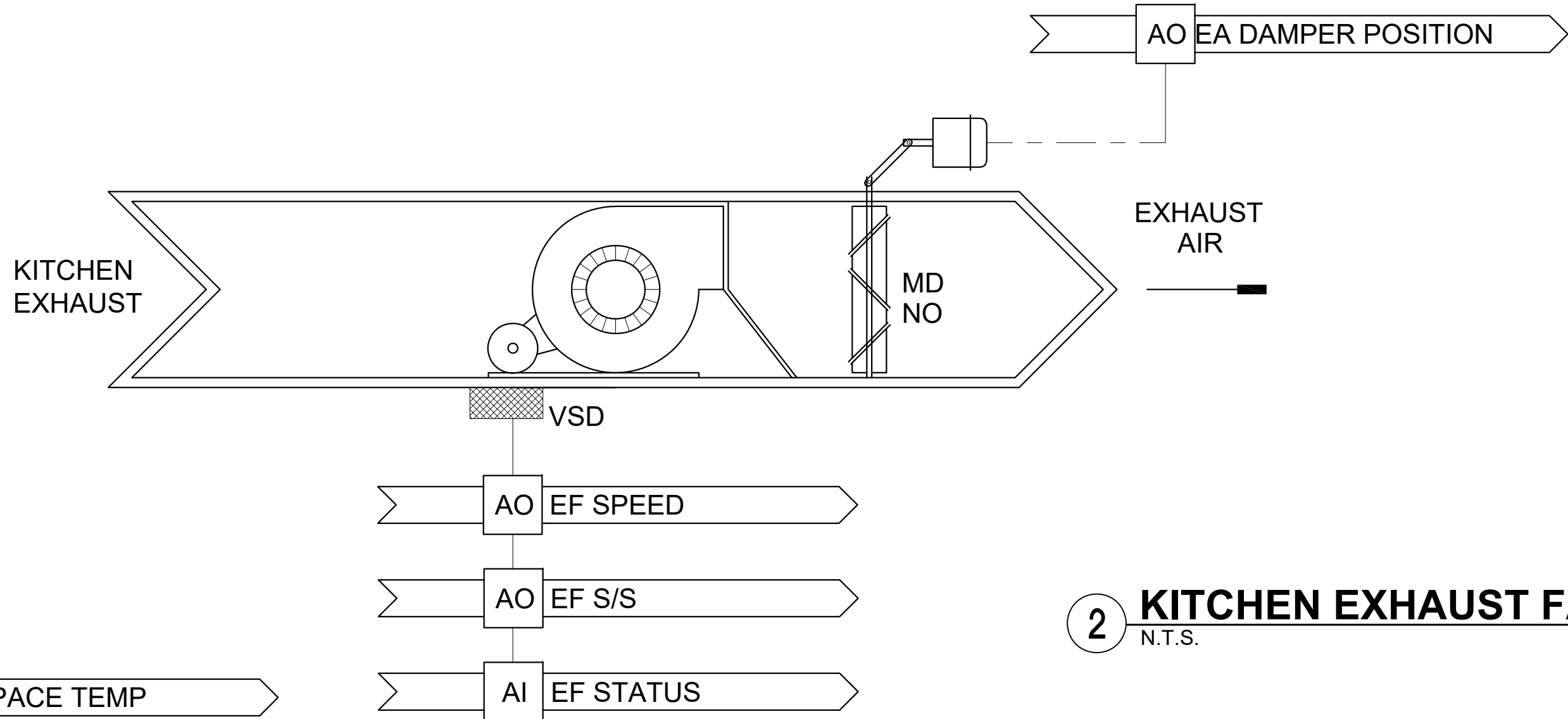


1 TWO-PIPE FAN COIL UNIT CONTROLS

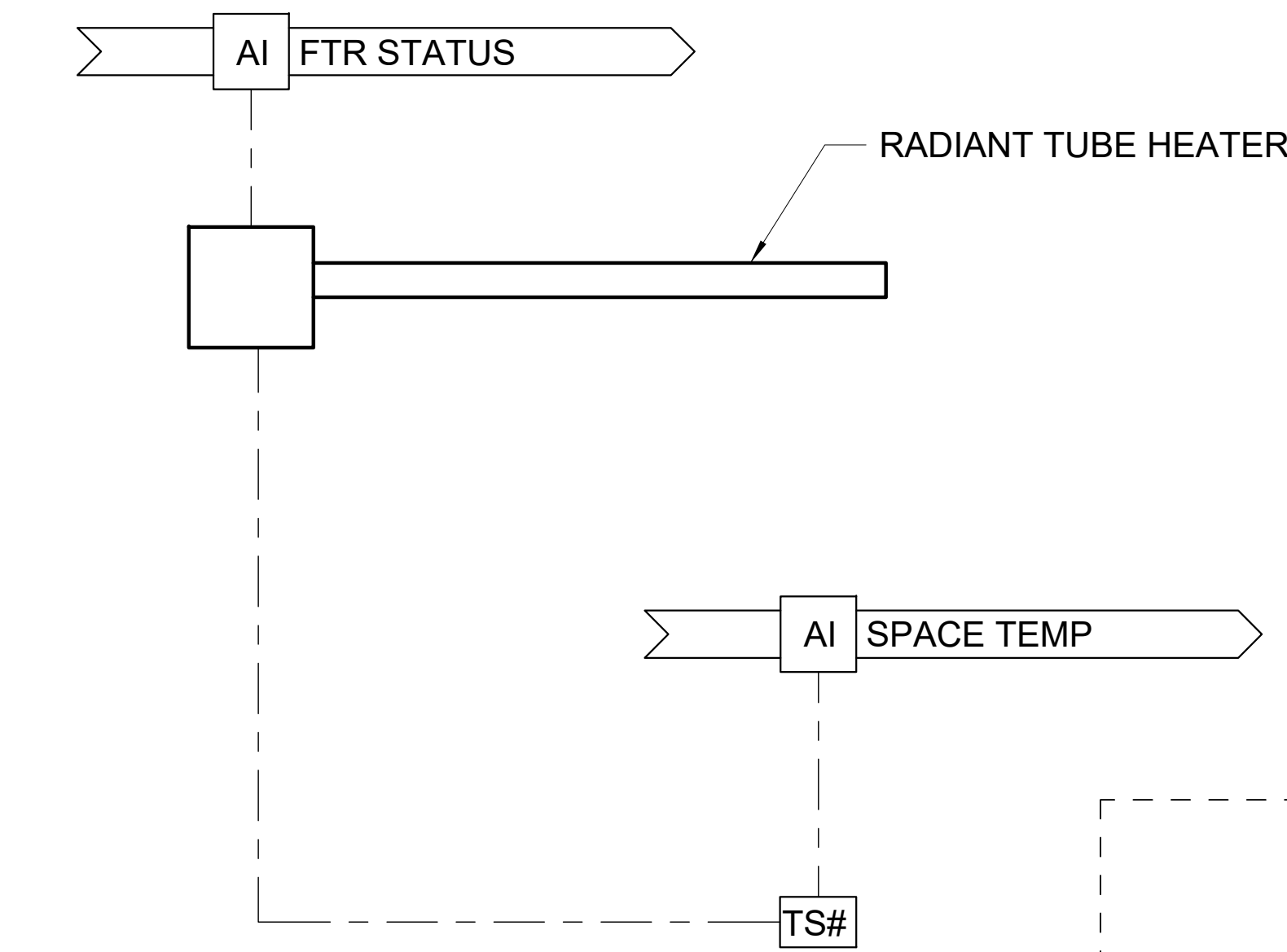
POINTS LIST (PER EACH FAN)							
ADDRESS	POINT DESCRIPTOR	POINT TYPE					REMARKS
		DI	AI	DO	AO	VP	
	EF S/S				■		
	EF STATUS		■				
	SPACE TEMP		■				
	EF SPEED				■		
	EA DAMPER POSITION				■		
	OA DAMPER POSITION				■		

SEQUENCE OF OPERATION

- WHEN SPACE IS OCCUPIED, THE INTAKE AIR DAMPER SHALL OPEN. ONCE DAMPER POSITION IS PROVEN, THE EXHAUST AIR DAMPER SHALL OPEN AND THE EXHAUST FAN SHALL RUN.
 - WHEN THE SPACE TEMPERATURE IS ABOVE 80F (ADJ.), THE FAN SHALL RUN AT HIGH SPEED.
 - WHEN THE SPACE TEMPERATURE IS BELOW 80F (ADJ.), THE FAN SHALL RUN AT LOW SPEED
- WHEN SPACE IS UNOCCUPIED, FAN SHALL SHUT OFF AND DAMPERS SHALL CLOSE.



2 KITCHEN EXHAUST FAN: E - 2

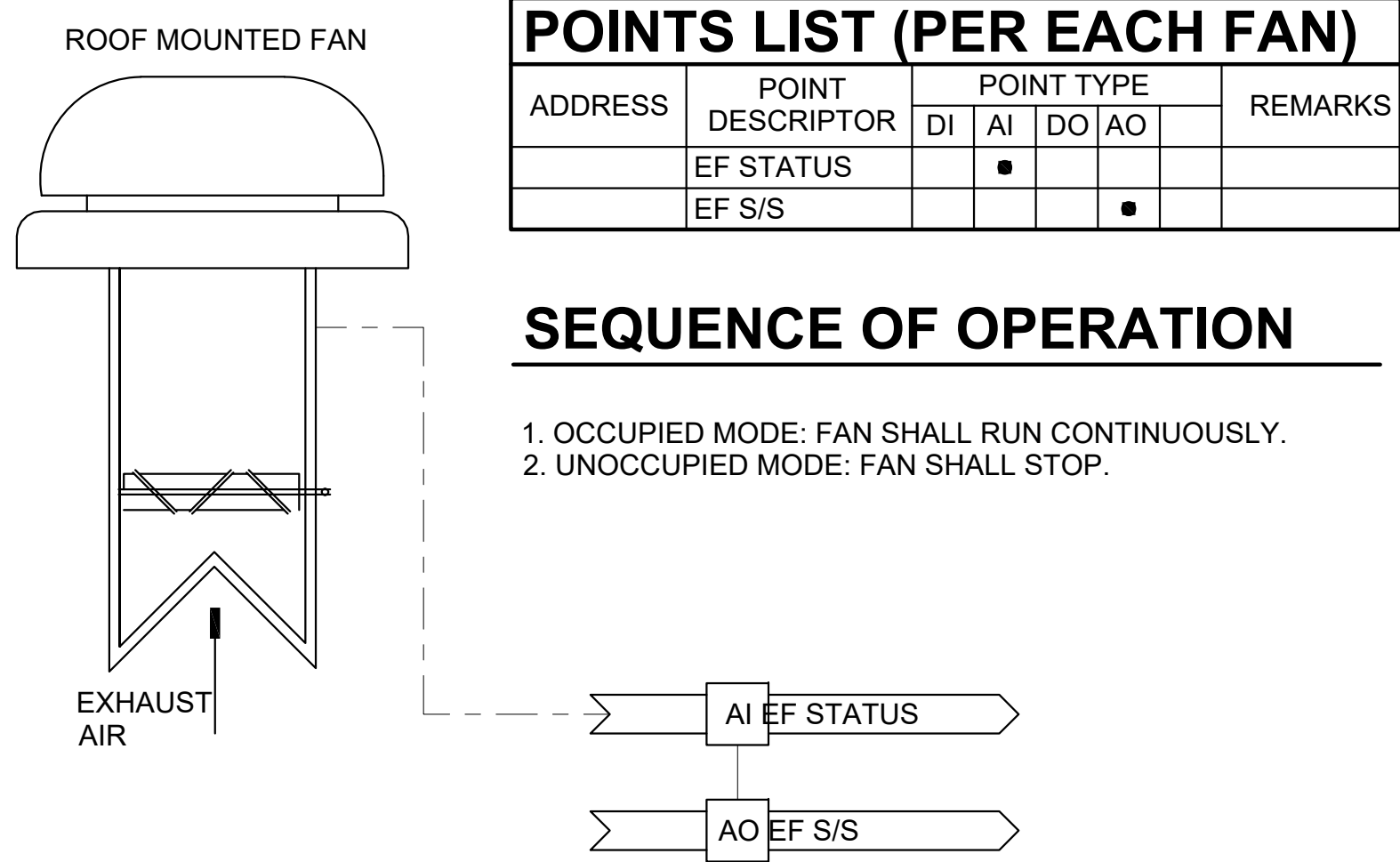


3 FIN TUBE RADIATOR

POINTS LIST						
ADDRESS	POINT DESCRIPTOR	POINT TYPE				REMARKS
		DI	AI	DO	AO	
	SPACE TEMP					
	HTR STATUS					

SEQUENCE OF OPERATION

- BELOW 55 F OUTSIDE AIR TEMPERATURE, RADIANT TUBE HEATER SHALL MAINTAIN 55 F (ADJ.).
- ABOVE 55 F OUTSIDE AIR TEMPERATURE, THE UNIT SHALL BE OFF.



2 ROOF-MOUNTED EXHAUST FAN: RV-16

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Revisions:

Date

Revisions Continued:

Date

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Defining EXCELLENCE in the 21st Century

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820 S. Damen Ave.

Chicago, IL, 60612

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Office of Construction and Facilities Management

VA

U.S. Department of Veterans Affairs

Drawing Title

TEMPERATURE CONTROL NOTES II

Approved Project Director

Phase

BID DOCUMENTS

FULLY SPRINKLERED

Project Title

AHU-AC21 REPLACEMENT SERVING NUTRITION FOOD AND SERVICE

Location

JESSE BROWN VAMC

Date:

October 30, 2025

Checked:

D. Moreau

Prepared:

A. Nguyen

Project Number

537-26-200

Building Number

1-B

Drawing Number

TC-2

Dwg. 15 of 16

Office of Planning and Construction

Department of Veterans Affairs

A

three inches = one foot

B

one and one half inches = one foot

C

one inch = one foot

D

three quarters inch = one foot

E

one half inch = one foot

F

three eighths inch = one foot

G

one quarter inch = one foot

H

one eighth inch = one foot

I

one eighth inch = one foot

J

one eighth inch = one foot

K

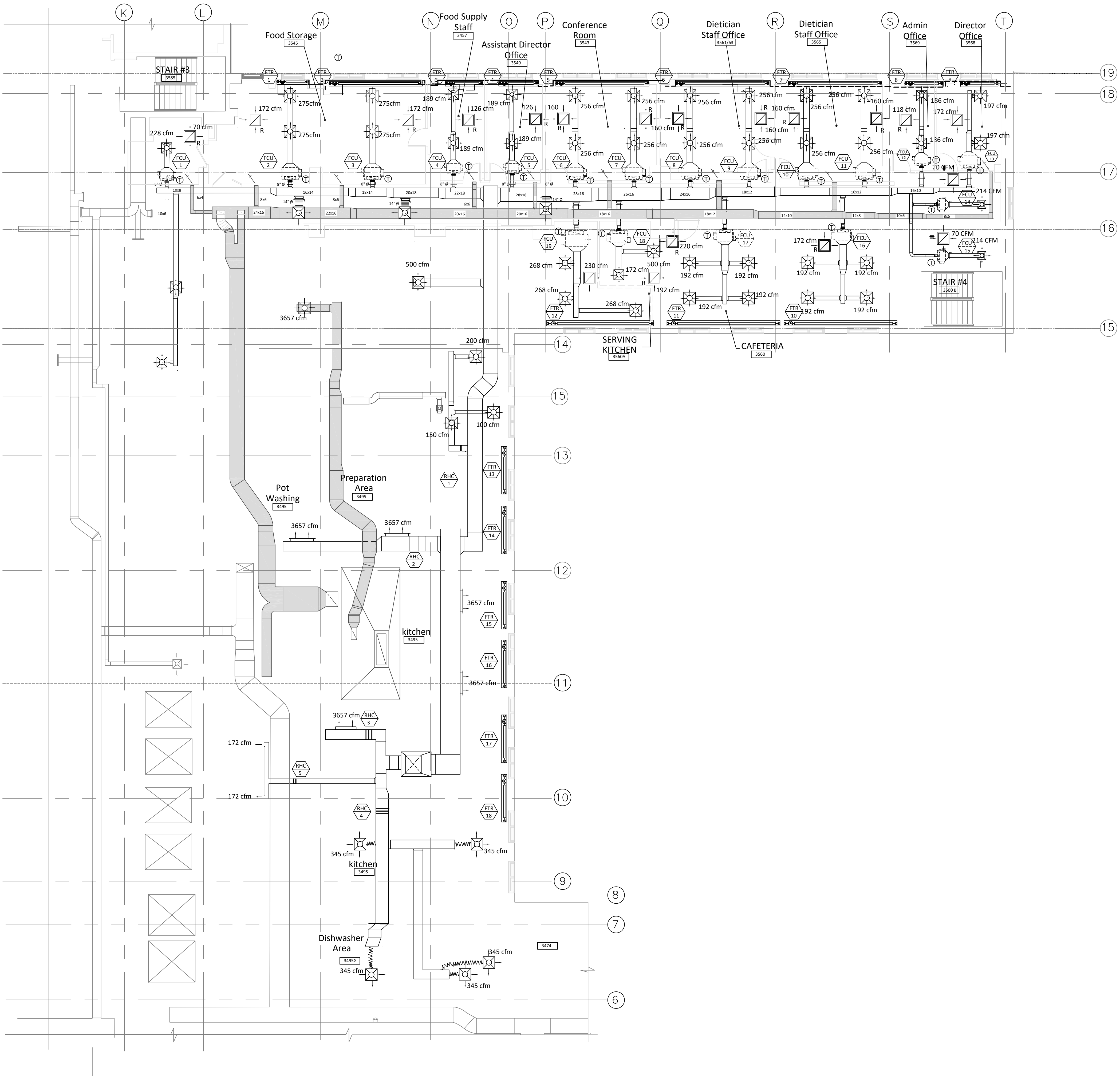
one eighth inch = one foot

L

one eighth inch = one foot

M

one eighth inch = one foot



VENTILATION FLOW RATES - BLDG 1-B, 3RD FLOOR
SCALE: 1/8" = 1'-0"

- GENERAL NOTES**
- CONTRACTOR SHALL MAKE USE OF THE EXISTING HOT WATER PIPING TO AND FROM THE REHEAT COILS. CONTRACTOR SHALL REPLACE THE EXISTING REHEAT CONTROL AND ISOLATION VALVES.
 - SINCE THE INSTALLATION IS CALLING FOR NEW TWO-PIPE FAN COIL UNITS, THE CONTRACTOR SHALL MAKE USE OF THE EXISTING PIPING TO THE UNITS. PIPING HAS TO BE FLUSHED BEFORE CONNECTING TO THE NEW EQUIPMENT.
 - EXISTING FLEXIBLE CONNECTIONS TO THE FAN COIL UNITS SHALL NOT BE REUSED.
 - MISSING BALANCING VENTILATION DAMPERS SHALL BE INSTALLED.
 - ALL EXISTING SUPPLY AND RETURN DIFFUSERS SHALL BE REUSED.
 - DO NOT INTERRUPT EXISTING UTILITIES SERVICING OCCUPIED OR USED FACILITIES, EXCEPT WHEN APPROVED IN WRITING BY THE COR AND THEN ONLY AFTER TEMPORARY UTILITY SERVICES HAVE BEEN APPROVED AND PROVIDED.
 - CONTRACTOR MUST TEST & BALANCE THE INSTALLATION AS PART OF THE PROJECT SUBSTANTIAL COMPLETION AND MAKE ALL NECESSARY FLOW ADJUSTMENTS.
 - EXISTING POWER FEEDS TO THE NEW FAN COIL UNITS SHALL BE REUSED.

NOTE: CONTRACTOR SHALL OBTAIN AND VERIFY ALL DIMENSIONS AND CONDITIONS AT JOB SITE AND BE FULLY RESPONSIBLE FOR SAME. BRIDGING DOCUMENTS, DRAWINGS AND NARRATIVES ARE PROVIDED FOR DESIGN INTENT. THE DESIGN-BUILD ENTITY IS RESPONSIBLE FOR THE COMPLETE DESIGN OF A PROJECT THAT ADHERES TO ALL SCOPE OF WORK REQUIREMENTS, CODES, STATE AND FEDERAL REGULATIONS AND GUIDELINES.

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FULLY SPRINKLERED